

Government Involvement in a Mandated Insurance Market: The Case of Workers' Compensation Insurance

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Abstract

I exploit the formation of several new public options in workers' compensation insurance in the 1990s to study the development of insurer competition and underwriting performance, which help provide evidence to the debate about the productivity of forming a public option. I show that workers' compensation marketplaces became more concentrated with public options gaining market share. This result appears not to be driven by the public option taking only bad risks. The average underwriting profitability of insurers improved with the formation of the public options, but the evidence suggests that was primarily due to cuts in states' legislated benefit levels that coincided with the formation of public options and not the public options themselves. I also document that, while public options tend to charge higher prices than their private competitors, they may help to stabilize private insurer loss experiences and encourage efficient market segmentation.

Keywords: workers' compensation, public option, insurance markets, social insurance

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1 Introduction

How governments regulate markets takes many forms, from price controls to quality standards to direct government competition with the private market. This competition mechanism is usually a “public option,” whereby the government provides people with a publicly-funded option alongside private alternatives (like public libraries, pools, and schools). This is the first paper to provide empirical evidence as to the role of creating a public option in the US workers’ compensation insurance market, where some public options have existed for more than a century and others were formed within the last 30 years.

Workers’ compensation insurance pays benefits (the duration and extent of which are set by statute in each state) when a worker is injured or becomes ill from work-related conditions, providing medical coverage and a portion of lost income. There is notable variation in the structure of workers’ compensation insurance markets across the US. In some states, the government is the only seller; in others, only privately-owned insurers sell coverage. Yet, in other states, there is a mixture of these two approaches via a public option that makes workers’ compensation a unique line of business in US insurance markets. These public options (also known as “state funds”) compete with private insurers for non-residual risks.¹ Currently, 28 states and DC have private-only settings; four states use government monopoly designs. In the 18 states that use public options in competition with the private market, the role of public options is notable: they paid nearly \$5 billion in workers’ compensation benefits in 2021 (Welch et al., 2024). Over time, many states have changed their market involvement via the public option, by either creating one for the first time or privatizing a currently-operating one. After documenting historical market statuses in each state, I focus on several new public option formations in the 1990s.

Given that market failures² are typically offered as the motivation for forming public

¹A residual risk, by contrast, would be an employer deemed too risky to price adequately by the private market. Generally, these residual risks end up in a residual pool, which has its losses shared by all workers’ compensation insurers in the state.

²Government involvement in US insurance markets began with workers’ compensation (Greene, 1976). As Greene (1976) laid out, the rationale typically given is that the government ought to be involved in insurance because of residual market needs (i.e., the private market is not making coverage available), convenience,

options, the ultimate guiding question here is: was forming public options a “good” decision and, if so, for which stakeholder(s)? While this is not directly answerable, one way to get at the idea is to ask about the effects of public options in workers’ compensation insurance markets along various dimensions. In that spirit, I study three more specific research questions that can present a picture of what forming a competitive public option does in this context, informing various arguments about whether ultimate formation was beneficial. First, what role do public options play in affecting competition among insurers in the workers’ compensation marketplaces in which they enter? Second, does the public option play a role in improving the underwriting profitability (i.e., how much an insurer pays out in losses relative to what it earns in premiums) of the market? Third, do public options have a role in managing market prices, segmentation, or private insurer loss volatility?

To answer these questions and propose a new measure of workers’ compensation price per dollar of payroll, I use detailed, insurer-level data from the National Association of Insurance Commissioners (NAIC) spanning 1988-2020, supplemented with state-level benefit and payroll data from the US Census Bureau’s County Business Patterns series, the *Social Security Bulletin*, and the National Academy of Social Insurance. Throughout, I use states that maintained consistently private-only markets (i.e., no public option) as the comparison for states that formed new public options, given the structure of both sets of states before forming a public option was the same. This is in keeping with the ideal experiment, where a public option is formed as a single policy lever. In reality, however, many states legislated benefit cuts for all workers around the same time as a broader set of reforms. To deal with this confound, I match new public option states to private-only states based on drops in benefit levels in the same year as public option formation, which allows me to compare matched and unmatched results to better explore the unique role of the public option.

Relative to the first research question focused on market competition, I document

compulsory purchase, efficiency, or collateral social purposes. In the workers’ compensation argument for competitive public options, the reasoning is typically efficiency related or due to affordability and availability issues (Thomason et al., 2001).

that public option states experienced elevated concentration levels (as measured by the Herfindahl-Hirschman Index, or HHI, which increases when the market is more concentrated) before public option creation. After the public option begins operating, those states see an even larger increase in concentration, ultimately never returning to the levels of the private-only counterfactual. The results indicate a near doubling of the HHI, with most new public options taking significant market share. While a public option may have been expected to negatively disturb concentration and encourage competition (Greene, 1976; Atkinson and Stiglitz, 1980; Polyakova, 2016), I ultimately show that the high concentration levels that developed just after formation have largely increased over time, including through 2020. These results are unchanged by matching on simultaneous legislated benefit cuts and are unexplained by either 1) the acceptance of “bad risks” as a residual market mechanism or 2) differential changes in private insurer entry/exit.

With respect to insurer performance, I show that insurer underwriting profitability improves dramatically after public options commenced operation. Statewide average loss ratios were significantly higher (more than ten percentage points higher) in public option states before creating the public option than in private-only states. In fact, they were above one for several years, an unsustainable proposition for insurers who are operating at a loss before including their operating expenses. After the public options joined the market, loss ratios fell by about 20%.

Yet, matching on simultaneous benefit cuts erases much of the loss ratio improvement. This implies that the public option did not meaningfully affect average loss ratios by itself. While the matched event study results indicate statistically insignificant loss ratio coefficients after public option creation, they are broadly still negative. To explore some rationales for this proposition, I show that differential trends toward safer industries in public option states and the shifting of comparably worse risks to the public option (improving private insurer experiences) do not explain any improvement in loss ratios. Price reductions that are much smaller than legislated benefit cuts in public option states (i.e., price reductions more than

50% smaller than benefit reductions), even after matching on relative changes in benefit levels, still stand to explain improvements in average loss ratios in the long-term.

In the face of public options being related to distinct increases in concentration and no real improvement in insurer underwriting profitability, what role exists for a public option in the market? This question is especially important given I show - through 2020, after more than 20 years of operation for these new public options - that they have continued to perform as well as, if not better than, their private market competitors and take up a large market presence (with average market share over 30%). From the consumer's perspective (the employer, in this case), one rationalization of a public option's continued role is to keep prices down. To get at this point, I propose a new standardized price measure in workers' compensation insurance that mitigates traditional insurance economics issues with observing the product of price and quantity (premiums) but not the individual elements. This price measure can be created for the entire market or for just the private market or public options.

I find that public options in these new public option states charge very similar, and almost always slightly higher, prices relative to their private insurer counterparts. The evidence suggests that the public option allows some segmentation in the market, whereby public options underwrite employers in higher risk industries (and charge them as such, using robust experience-rating, as is common in workers' compensation) and allow the private market to charge lower prices on average. These higher risk industry policies do not need to be "residual" risks for this to be true.³ From the perspective of private insurers, one other role of a public option could be helping to improve market stability (loss ratios at the insurer level). I provide some evidence that public options improve stability on average, but it is not true in every state. The roles for continued public option presence in this market appear to be promising avenues for future research.

This study contributes to the literature in a few important ways. Naturally, this study

³In fact, if public options charge higher prices and maintain lower loss ratios than the private market (which I show to be true), then the higher risks I refer to may not be "residual" at all. If they were, prices would not be able to match risk perfectly (hence the push out of the regular market and into the residual market).

adds to the broad literature on workers’ compensation insurance. In light of recent work outlining the lack of justification for mandates for workers’ compensation insurance coverage (Cabral et al., 2022), policymakers may consider turning to the design of Texas’ system, which has no coverage mandate and a public option. Importantly for state-based policymakers considering these changes, understanding what the impact of the public option is on the workers’ compensation market helps isolate one key policy lever. In the workers’ compensation literature, this would be the first study to apply robust causal inference strategies to learn about public options in workers’ compensation, not just in the years of creation but over the long term, with more than 30 years of detailed, insurer-level data. There has been ample research on workers’ compensation insurance markets over time, but it has focused on the markets without public options (for example: Carroll (1993); Danzon and Harrington (1998, 2001); Harrington and Danzon (2000)), the cost efficiency differences between private and public insurers (for example: Butler and Worrall (1986); Thomason et al. (2001)), the moral hazard effects of benefits themselves (see Meyer et al. (1995) for just one example), or spillovers between voluntary and residual markets (Kwon and Grace, 1996).

This paper also contributes more broadly to the literature on insurance markets beyond just workers’ compensation; this is the first study to empirically estimate the effect of a public option in US insurance markets.⁴ Public options have been widely studied in other US contexts. One of the most common areas to study a public option in the US is in education, where Dinerstein and Smith (2021) provide recent evidence in this arena and review prior work. Other examples exist, too. Caskey (2010) analyzes off-street parking in Philadelphia, Levitin and Wachter (2013) discuss household finance and mortgages, Hart et al. (1997) discuss public provision of prisons and garbage collection, and Amirkhanyan et al. (2008) study nursing homes.

In US insurance markets, researchers have generally focused on understanding what a

⁴These effects have been studied outside of the US, however. Work has studied various settings, including health insurance in Germany (Polyakova, 2016), pharmacies in Chile (Atal et al., 2024), and, in Sweden, ambulance systems (Knutsson and Tyrefors, 2022) and elderly care centers (Bergman et al., 2016).

public option could do in the health insurance context (Cebul et al., 2011; Miller and Yeo, 2019; Craig, 2022), including using “Medicare-for-All” (Shepard et al., 2020), or focusing on certain state-based arrangements that look like public options but function in (importantly) different ways. For instance, Washington recently implemented an approach where private insurers could sell a “public option plan” with regulated features (Sen et al., 2022). Colorado took a similar approach with running public option plans through private insurers but mandated that private carriers offer one (Kumar and Adashi, 2023). Nevada is set to implement a similar approach by 2026. All fall well short of the competitive public option design present in workers’ compensation, where there is no public-private coordination, and as has been called for in the national health insurance setting.⁵ While workers’ compensation insurance and health insurance are different products among various dimensions, they are also similar in that they are largely employer-provided, highly regulated, and provide medical benefits. In this way, this paper may also lend some insight to creating a competitive public option in health insurance markets. There are also other insurance arrangements that have public options but are yet to be studied (to the best of my knowledge), namely mandatory non-occupational disability insurance (Greene, 1976).

This paper also contributes to the literature on pricing in US insurance markets, specifically related to residual markets and workers’ compensation, given that most public options in the workers’ compensation setting are “insurers of last resort” (which is similar to providing guaranteed issue in the health context). There is a literature studying residual market mechanisms in other property-casualty insurance markets: auto insurance (for just a few examples, see: Bajtelsmit and Bouzouita (1998); Browne and Wells (1999); Grace et al. (2013)) and property insurance (Born et al. (2021) provide an extensive review of the related work). It has also been discussed in the health context (Finkelstein, 2004). Taking advantage of

⁵There has been ample attention paid to public versus private provision of insurance in terms of traditional Medicare (public provision) and Medicare advantage (private provision) plans (Curto et al., 2019). Importantly, that setting differs from the pure public option, given the government provides set payment amounts to private carriers for each enrollee. My analysis below focuses on public options in active competition, not coordination, with the private market.

the unique nature of workers’ compensation (legislated benefit levels are constant across employers within a state with nearly universal coverage of employees), I propose a novel price measure that combines several data sources to produce a unit price (i.e., per \$1 of payroll).⁶ This unit price measure is new to the workers’ compensation space (Carroll and Kaestner, 1995; Kwon and Grace, 1996; Thomason et al., 2001; Danzon and Harrington, 2001) and is a new concept in the sense that I can provide prices specific to public options, contributing to the residual market literature broadly.⁷

The remainder of the paper is organized as follows. Section 2 describes the historical and institutional background of workers’ compensation. Section 3 describes the data, while Section 4 provides a descriptive explanation of what occurred at times of public option formation. A discussion of simultaneous legislated benefit cuts follows in Section 5, which also presents the empirical methodology. Sections 6, 7, 8, and 9 present results, discuss prices and mechanisms, consider roles for public options in marketplaces, and conclude, respectively.

2 Institutional Background

In the early 1900s, the fight for workers’ compensation insurance was just getting underway. With no federal movement, it largely became a state issue. Several debates surfaced about how to best operationalize workers’ compensation programs, with a more niche debate erupting in many states over how to provide workers’ compensation insurance. Meaning, should it be provided with government involvement, either in competition with the private market or in a monopoly arrangement? This debate featured disagreements between organized labor (desiring state monopolies), insurers (desiring no state involvement), and cautious

⁶In most insurance contexts, premiums are observed but are a function of both price and quantity, and there does not exist a way to separate that out in insurance data. In workers’ compensation, however, the quantity space is more well-known - almost all workers are covered and the amount of coverage provided to each employee is constant within a state in a given year. The measure I propose also takes into consideration several challenges with using traditional insurance price proxies (i.e., based on the loss ratio) in workers’ compensation (Thomason et al., 2001).

⁷The price measure I propose is an extension of Carroll and Kaestner (1995) using premiums per non-agricultural employee. Their price measure was not differentiated between private and public option insurers to allow for a residual market extension.

employers (not wanting poor state control to lead to higher costs and adverse selection issues) that led to heterogeneity across states that has persisted to this day (Clayman, 1959; Weinstein, 1967; Fishback and Kantor, 1996, 1998).

Before discussing the state variation in workers’ compensation insurance market formation in detail below, it is useful to provide a brief idea of what the system does for workers. Workers’ compensation provides almost guaranteed payment to employees for injuries at work (thus, it is an occupational disability program).⁸ Yet, it is also the exclusive remedy available to employees (employees cannot sue their employers), functioning as a strict liability system. As long as an employee injury satisfies a few conditions (harm caused by an accident arising out of, and in the course of, employment), they will be paid for some lost income and all relevant, covered medical expenses. Without a workers’ compensation insurance program, typical negligence rules would apply, whereby the employer and employee must resolve the issue in court (or settle out of court), with more risk – both in who wins and how much is paid. Coverage for workers’ compensation is provided by the employer for its employees; workers’ compensation is not a product available for employees to purchase directly. Benefits are prescribed by the state, so each firm is purchasing the same amount of coverage per employee.

2.1 Workers’ Compensation Insurance Marketplaces

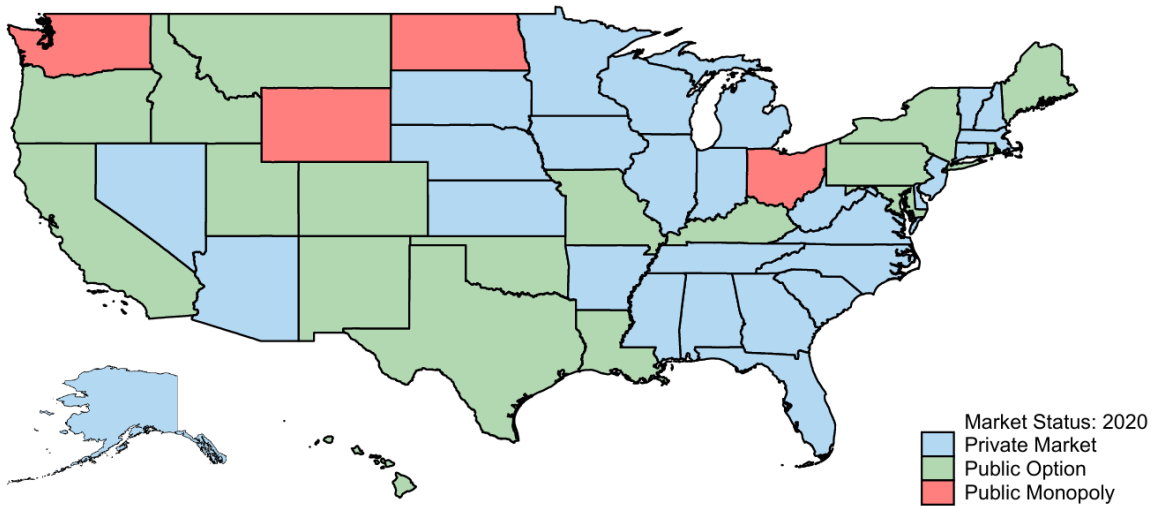
The structure of the workers’ compensation insurance marketplace varies from state to state (and over time, as will be described below). To group state⁹ marketplaces, I start with thinking about how employers are permitted to provide workers’ compensation coverage to their employees, while retaining the exclusive remedy nature: either through buying insurance or arranging self-insurance. Within the insurance path, there are three broad categories that define the US workers’ compensation insurance market: 1) private market

⁸This does not mean non-occupational injuries or illnesses do not find their ways into workers’ compensation claims. Smith (1990) documents the prevalence of increased claims on Mondays that depict evidence of concealment and delayed treatment.

⁹Here and throughout the paper, I refer to states and jurisdictions interchangeably and include the District of Columbia (DC) in those discussions - which is why I reference 51 “states” or jurisdictions. I do not reference any US territories in this paper, though they each have their own market structure.

states, 2) public option states, and 3) public monopoly states. Figure 1 provides the state of the US marketplace in 2020, the last year of my data.¹⁰

Figure 1: Workers' Compensation Market Structure in 2020



Notes: This figure displays broad workers' compensation insurance market structure as of 2020. Sources: National Academy of Social Insurance annual reports and author's own statutory & policy research. Sources for this statutory and policy research are compiled and available upon request.

In private market states, the only source of coverage for employers comes via the private market. In these settings, the government usually sets up a residual market mechanism (often a residual pool) which reinforces the idea of compulsory coverage for those employers who cannot get coverage on their own. There are 28 states plus DC in this category. As a second category, 18 states utilize a public option marketplace. In these settings, the marketplace is similar to a private market but where an active competitor is the public option insurer, which was created by the government and typically operates like a mutual insurance company. Finally, there are monopoly states, of which there are four, wherein the only source of coverage is via the government.¹¹

¹⁰This map also applies through 2024, given there have been no market changes since 2020.

¹¹All public option and private market states allow for employers to become certified self-insurers. Among public monopoly states, North Dakota and Wyoming do not allow self-insurance, while Ohio and Washington do.

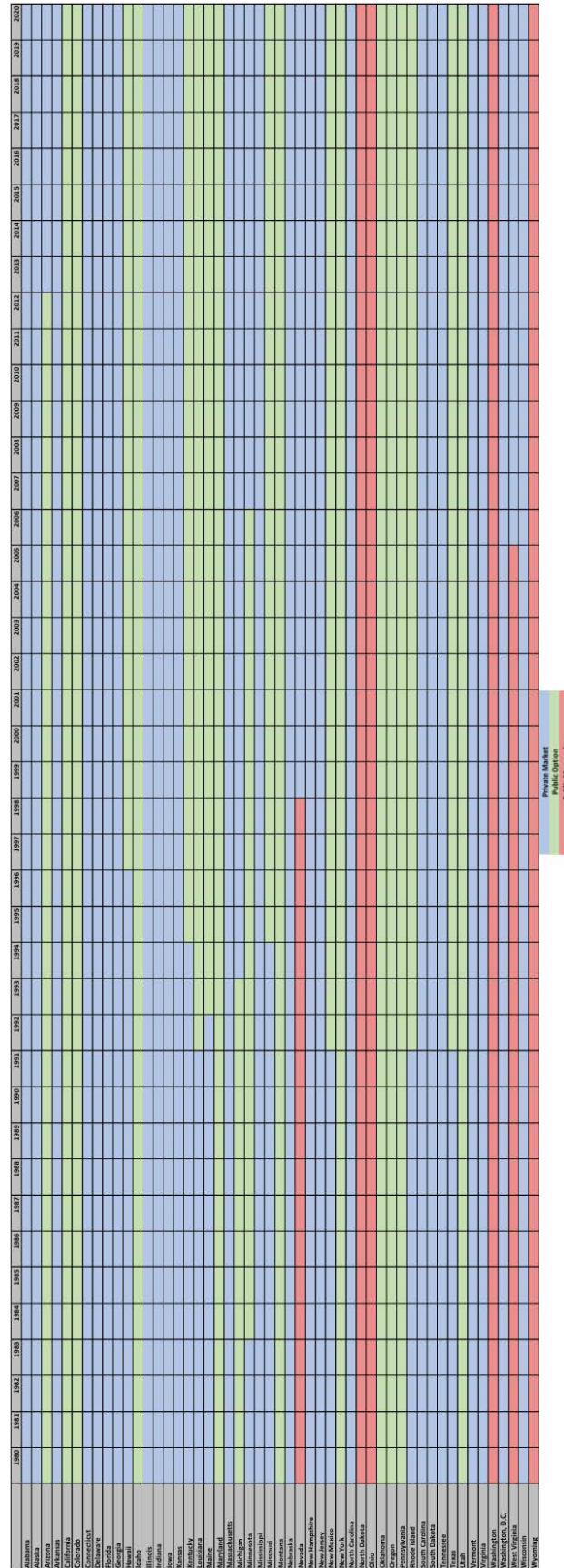
There is additional variation in public option states, most notably related to residual market responsibilities and federal tax payments. Four of the eighteen states with a public option - Idaho, Missouri, New Mexico, and Oregon - do not require that the public option accept all risks. Meaning, the public option does not serve as the residual market mechanism. In the other fourteen states with public options, the public option does serve as the residual market. Within the fourteen public options that are also the residual market mechanism, two of those states - Maine and Utah - no longer exempt the public option from federal taxation. While this means that these two states are no longer classified as public option states strictly by the National Academy of Social Insurance, the forthcoming analysis does not use Utah and will only assess Maine's public option when it was still exempt from federal taxation (before 2007).

2.2 Changes to Market Structures Over Time

While Figure 1 provides a snapshot of the cross-sectional variation in state workers' compensation insurance markets in a single year, Figure 2 provides a mapping of state marketplaces from 1980 through 2020. The color discontinuities depict structural changes to the marketplace. For instance, Nevada went from a public monopoly (in red) to a private market (in blue) in 1999. The same happened in West Virginia in 2007. This graphic not only helps identify the time of changes, but it highlights just how much variation there is across the US, by state and over time. It is also noteworthy that many states did not make changes over this period. In total, there were nine states that created a public option over this time period: Minnesota (1984), Louisiana (1992), New Mexico (1992), Rhode Island (1992), Texas (1992), Maine (1993), Kentucky (1995), Missouri (1995), and Hawaii (1997). While Minnesota had completely privatized its public option by 2008, the other eight states continue to have public options.¹²

¹²Michigan and Arizona also privatized their public options during the study period - Michigan in 1994 and Arizona in 2013. Missouri plans to do so in 2025.

Figure 2: Workers' Compensation Market Structure Changes, 1980-2020

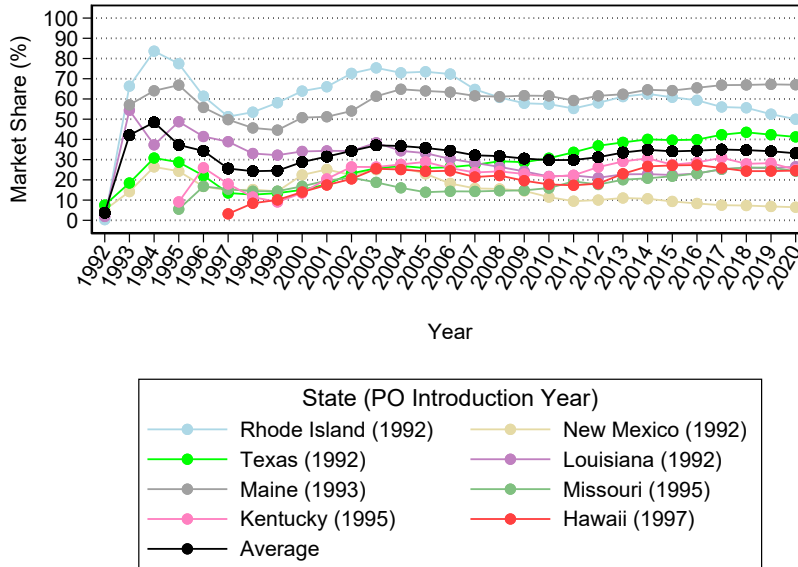


Notes: This figure displays broad workers' compensation insurance market structures from 1980 through 2020. Blue corresponds to private market states, while green corresponds to public option states and red to public monopoly states. Sources: National Academy of Social Insurance annual reports and author's own statutory & policy research. Sources for this statutory and policy research are compiled and available upon request.

2.3 Public Option Relevance

Among the eight states that created a new public option without reversal, most of which I use as the main set of focus states in this paper, public options have maintained a large role in the market over time. Figure 3 shows the market share of the public option from 1992 to 2020. Market share is defined as the percent of total direct premiums in the workers' compensation market written by the public option.

Figure 3: Public Option Market (PO) Share by State



Notes: This figure displays the market share of each public option based on the portion of direct premiums written in the state, as given by the colors in the key along with their years of commencement. The average market share across the eight new public options is also provided over time. Sources: NAIC 1988-2020.

After a slow first year or two after formation, they eventually develop a notable presence that lasts through the 20-year anniversaries of their formations. Every new public option experienced market shares of at least 10% within two years of forming. Over time, the market share of these public options persisted: market shares exceeded 20% in all but one of the new public option states in 2020. In three states, public option market share was more than 40%, with the average above 30%.

3 Data

This paper employs data from numerous sources. I use detailed, insurer-level data aggregated to the state level to assess the effect of public option formation on competition and insurer loss ratios. I then use a panel of state payroll figures and payer-level information on workers' compensation benefits paid to match states based on benefit levels and to construct a price measure.

3.1 Insurer Data by State, Line of Business, and Year

I utilize data from the National Association of Insurance Commissioners (NAIC) over the 1988-2020 period. The data provide information on the losses and premiums of each insurer, by state and year. The public options that I investigate in this study started reporting data to the NAIC in the year of their formation.

3.2 Benefits and Payroll Data

Data on workers' compensation benefits payments come from the *Social Security Bulletin* (SSB) for the years 1988 to 1993 and from the National Academy of Social Insurance for the years 1994 to 2020. Both of these sources are largely available publicly.¹³ For each year and each state, I identify the amount of workers' compensation payments made by 1) private insurance carriers, 2) public options (state funds), and 3) self-insured employers. These total benefits paid numbers include medical and lost income benefits. Generally speaking, these data validate the discontinuities in Figure 2.¹⁴

¹³The following reports provide the data: Nelson (1992, 1993); Schmulowitz (1995); Mont et al. (2002); Sengupta et al. (2005, 2010, 2013, 2014); Sengupta and Baldwin (2015); Baldwin and McLaren (2016); McLaren and Baldwin (2017); McLaren et al. (2018); Weiss et al. (2019); Murphy et al. (2020, 2021); Murphy and Wolf (2022).

¹⁴The SSB publications mistakenly suggest public options in New Mexico and Rhode Island did not begin paying benefits when they actually did, so those two states are dropped from estimates that rely on benefits paid data because there is no easy way to know how much of the private carrier payments should be from public options in those years of incorrectly-reported non-existence. A note must be made about Hawaii, which was cited as having a public option in 1992 and 1993 despite the public option not being created until 1997 (Schmulowitz, 1995). This was an error not acknowledged by the SSB publications (as they ended with the 1993 year), but the Hawaii government noted in a separate report that these were from the state Special Compensation Fund (Pang et al., 1993) and the National Academy of Social Insurance did not report Hawaii state fund losses paid between 1994-1996. Following the National Academy of Social Insurance's process of prorating the special injury fund payments across privately-insured and self-insured losses, I take the (incorrectly reported) state fund losses paid value and add it to privately-insured losses

Annual employment, payroll, and firm (number of reporting units) data, at the industry-state level come from the Census’ County Business Patterns (CBP) series. With the available data, I need to assume all payroll is covered by workers’ compensation, which is largely true except for various small firm exemptions around the US. I cannot explicitly say how much of the CBP payroll is “covered,” though I can cite that more than 97% of payroll and employment was covered by workers’ compensation in 2015 (BLS, 2015; SSA, 2017). This number is very stable, with the most recent estimate suggesting 97.5% of all non-federal jobs reported via unemployment insurance filings were covered in 2021 (Welch et al., 2024).

4 Public Option Formation

In this section, I investigate the raw trends in market concentration and insurer underwriting profitability before and after the formation of the public option. In Section 5 below, I describe a stacked event study approach that formally tests for pre-trends and applies multi-level fixed effects. Figure 4 shows the 32 states (including DC) in my sample. The public option states are depicted in dark blue. I only include the eight states that formed a public option in the 1990s (without reversal). The comparison states, shown in light blue, are the states with a private market throughout the sample period. This is the useful comparison for this exercise given both groups had the same, private-only structure before new public options were formed. The sample does not include the nineteen states that either (1) had a public monopoly (four), (2) had a public option throughout the entire sample period (ten), or (3) made other changes to their public option (e.g., privatizing a public option or monopoly; five states did so).

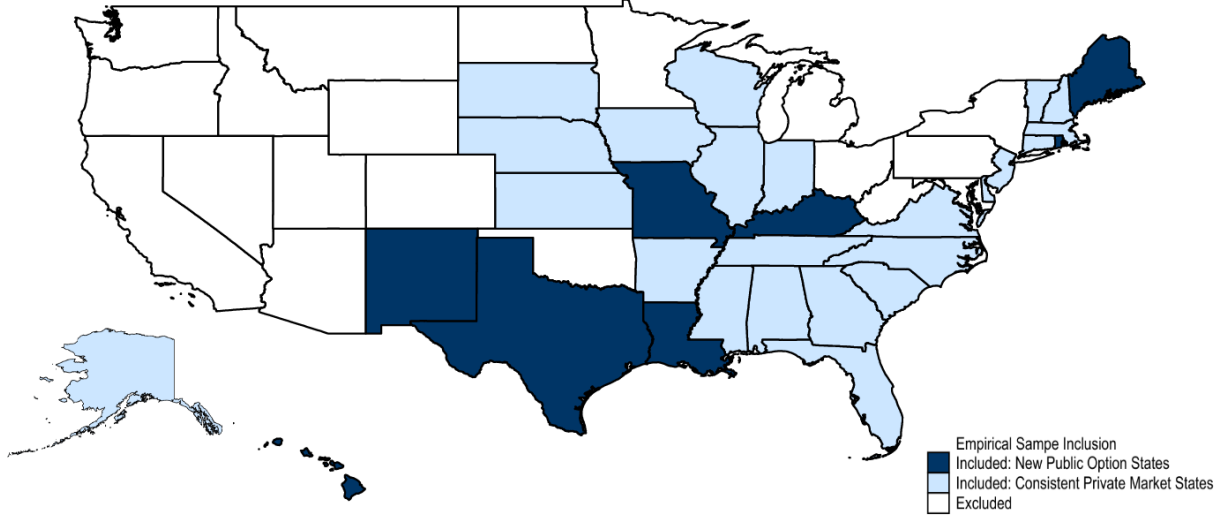
4.1 Market Competition: Concentration

The first outcome of interest focuses on the level of insurer competition in each state marketplace. To measure competition, I use the Herfindahl-Hirschman Index or HHI. The HHI is the sum of the squared market share of each insurer in a state.¹⁵ A higher HHI value

and self-insured losses, based on the percentage of total losses paid by private insurers and self-insured firms.

¹⁵This setup excludes insurers who report a negative DPW number, though the results are robust to including those firms. Appendix Table A.1 shows that less than 6% of insurer-year observations report negative DPW

Figure 4: Empirical Sample Inclusion



Notes: This figure displays state-level inclusion in the main empirical sample, as described in the text. States with no color are excluded; states with color are included (light blue states are consistently private market states; dark blue states are the new public option states).

means that more employers in the state are getting their insurance from fewer insurers.

Figure 5 presents trends in the median HHI value among the states that consistently used a private market in panel (a) and new public option states in panel (b).¹⁶ From panel (a), median HHI reaches a maximum of 0.036 in 1993 and a minimum of 0.019 in 2000. The small range documents the stability of insurer concentration in states with consistently private markets.¹⁷ The picture in panel (b), however, is quite different. Two things become immediately clear. First, the median HHI increases by about 30% across the pre-period, from 0.050 four years before formation ($t = -4$) to 0.064 in year $t = -1$. Second, while

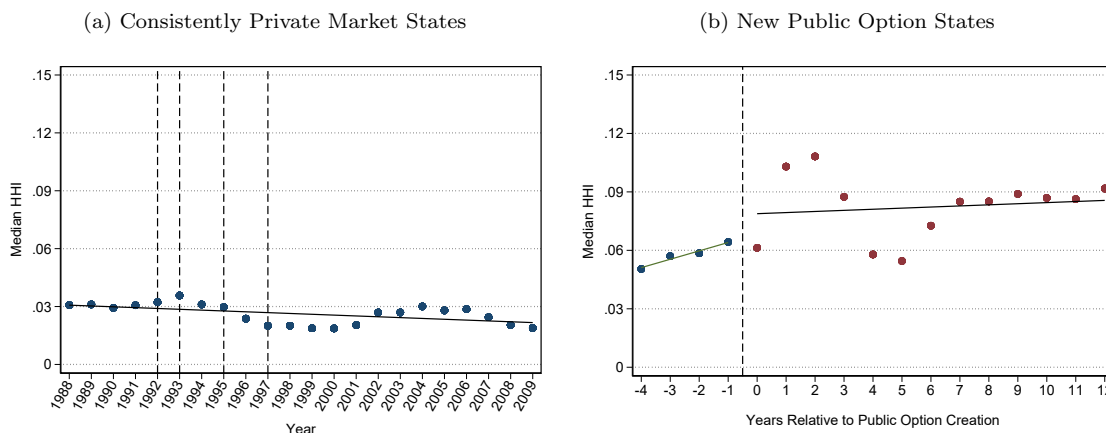
in workers' compensation. This table also compares negative versus non-negative firm-year observations. Appendix Table A.2 places firms into ever negative and never negative groups to provide comparisons. As is clear from those comparisons, negative DPW firm-year observations tend to come from smaller stock insurers.

¹⁶Appendix Figure A.1 uses the same setup but graphs means rather than medians. Using means, these trend differences are even more stark after a public option is formed.

¹⁷Note that panel (a) runs from 1988 (four years before the first public option was formed, which was in 1992) through 2009 (twelve years after the last one was formed, which was in 1997). This is intended to roughly mimic the event time in panel (b) for new public option states, given there is no "event" for states without a public option. For clarity, the four public option creation years are marked with vertical, dashed lines.

there is no noticeable shock to the median HHI when the public option joins the market at $t = 0$, median HHI values jump to 0.103 and 0.108 in years $t = +1$ and $t = +2$, respectively. This value in year $t = +2$ is more than double what it was at $t = -4$ (and nearly four times higher than in their private-only counterparts). While median HHI falls to 0.055 in year $t = +5$, it appears new public option states only become increasingly more concentrated in the long run, with a median HHI of 0.092 by year $t = +12$. As Figure 3 showed earlier, these trends in median HHIs align well with the increasing relevance of public options.

Figure 5: Trends in Concentration (HHI) - Medians



Notes: This figure displays median HHIs across states in consistently private market states in panel (a) and new public option states (b), across calendar time (year) and event time, respectively. A linear trend line is drawn through all points in panel (a) and drawn separately in panel (b) before and after public option formation (at time = 0). Vertical lines in panel (a) indicate public option creation years in public option states. Sources: NAIC 1988-2020.

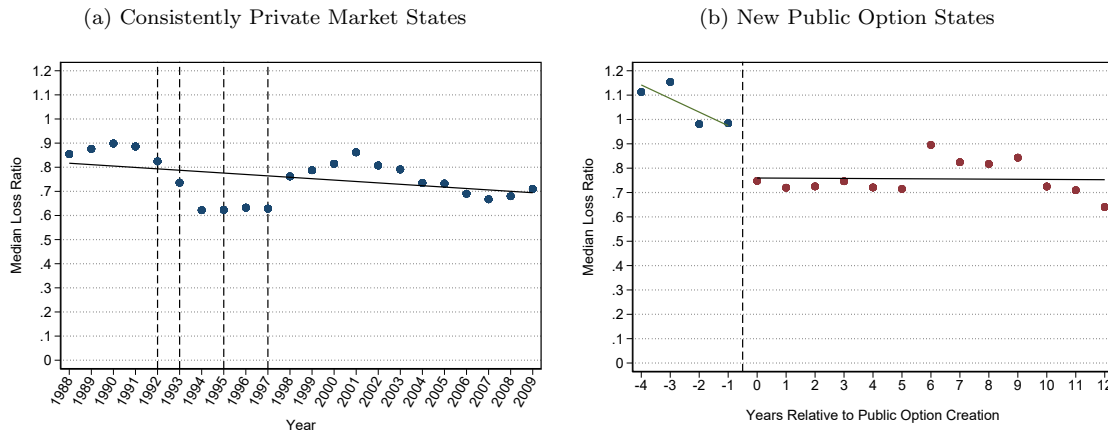
4.2 Underwriting Profitability: Loss Ratios

To investigate insurer underwriting profitability, I use the loss ratio, which is direct losses incurred over direct premiums earned. Each element is first aggregated across all insurers in a state in a given year. Loss ratios are one of the few measures of performance that can be built back to 1988 using the NAIC data, and they provide insight as to the underwriting performance of an insurer (i.e., the success of the insurer at predicting losses, charging premiums, and paying claims). Of course, this is an imperfect measure: it does not account

for expenses nor investment returns.¹⁸ Loss ratios are a measure of performance in the sense that, from an insurer's perspective, a lower loss ratio implies greater profitability. For instance, a loss ratio over one is not sustainable because, even without expenses included, the insurer is exhibiting issues matching their prices and quantities sold (premiums) to losses. This is especially true at the state level.

Figure 6 presents the raw trends in the median loss ratio among consistent private-only states in panel (a) and new public option states in panel (b).¹⁹ Loss ratios are decreasing in similar periods for both groups of states, which tends to be driven by having larger, faster decreases in losses paid (or expected to be paid) than premiums earned (see Figure A.3). In panel (a), the median loss ratio drops from a high of 0.899 (i.e., insurers were paying \$0.90 in losses for every \$1 collected in premiums) in 1990 to 0.622 by 1994. The median loss ratio picked back up to 0.862 in 2001 before steadily decreasing to 0.710 by 2009.

Figure 6: Trends in Loss Ratios - Medians



Notes: This figure displays median loss ratios for consistently private market states in panel (a) and new public options states in panel (b), across calendar time (year) and event time, respectively. A linear trend line is drawn through all points in panel (a) and drawn separately in panel (b) before and after formation (at time = 0). Vertical lines in the left-side graphs depict public option creation years. Sources: NAIC 1988-2020.

¹⁸Data constraints tied to the early years in which public options were formed mean that expenses and net investment gains are only available since 1991 and are provided at the insurer level (which would require assumptions about how to divide among the states and lines in which an insurer writes business).

¹⁹Trends in mean loss ratios can be seen in Appendix Figure A.2, and trends in the means of each loss ratio component can also be found in Appendix Figure A.3.

In panel (b), median loss ratios were much higher in the pre-formation period for states that adopted a new public option than at any point for consistently private states, ranging from 1.113 in year $t = -4$ to 0.985 in year $t = -1$. At the time of formation, there was an immediate drop, with a median loss ratio in year $t = 0$ of 0.748. Median loss ratios were fairly stable over the remainder of the period, before closing at 0.640. At first glance, it appears that public options were aligned with a steep improvement in the insurer experience via the loss ratio, given the nearly 25% drop at time $t = 0$. Yet, when comparing to the changes among the consistently private-only states in panel (a), there were also (slower) improvements in the loss ratio around the same time, which are seemingly independent from public option formations. These patterns are indicative of other legislative activity in states that never created a public option, which could also be relevant for public option states.

5 Legislated Benefit Levels & Empirical Methodology

Issues in the workers' compensation insurance market in the late 1980s and early 1990s were not limited to only the eight states that decided to create public options. Costs were a problem across the US. As a result, over half of state legislatures enacted major benefit cuts to both categories of benefits, income replacement and medical (Thomason et al., 2001). In fact, Thomason et al. (2001) report that the 1992-1998 period marked the first time in nearly 40 years that medical benefits experienced an annual percentage decrease nationally and that wage replacement benefits experienced an increase of less than 1%.

These types of changes to legislated benefit levels, while constant across all insurers in a state and year, could present challenges for simple comparisons between new public option states and consistently private-only states. This is especially true with a main outcome like loss ratios, where large changes in benefit cuts that are comparably larger in new public option states may have an out-sized influence on insurer performance if prices do not respond in the same fashion. Below, I first discuss the landscape of legislated benefit levels around the time of public option creation and then present an empirical methodology that may allow for better inference on a public option's *unique* impact on competition and insurer

experiences.

5.1 Workers' Compensation Benefits Paid

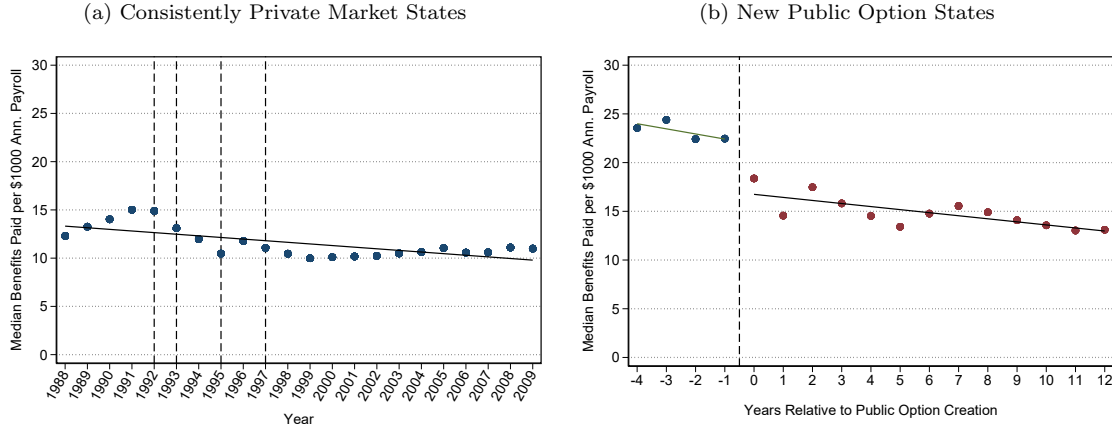
To track the legislated benefit levels, I make use of total workers' compensation benefits paid and payroll data described earlier. Due to data issues discussed above, two public option states are dropped from this setup (New Mexico and Rhode Island), but the remainder of the empirical sample is retained in these data. These benefits paid data largely track with legislated benefit changes in many states. As some evidence of that, Appendix Figures A.4 and A.5 show the benefits paid measure for the six new public option states and then the remainder of states (except the two that were dropped). The takeaway from that large array of graphs is twofold: 1) benefit cuts were quite substantial in new public option states, whereby public options did not simply take benefits payments one-to-one from private insurers (implying benefits must have decreased); and 2) most other states also experienced benefit cuts, though to varying degrees. These patterns are consistent with Thomason et al. (2001) and reporting of several large bills in various states.²⁰

Figure 7 provides the trends in median total workers' compensation benefits paid per \$1,000 of annual payroll, comparing the states with consistently private markets to the new public option states. Appendix Figure A.6 does the same for means. In addition, Appendix Figure A.7 shows each new public option state's total benefits paid amount over time, graphed against the private market average for reference.

The trends in the median (along with the referenced figures from the appendix) indicate that there were large relative benefit cuts in both groups of states, but most importantly,

²⁰Several of the more notable states in terms of benefit changes made legislative updates that are reflected in the data from the SSB and National Academy of Social Insurance, enhancing the trust in those data as (roughly) tracking of legislated benefit levels. For example, the data show Alaska as having experienced large benefit drops in or around 1988. In fact, Alaska legislated major reforms to its programs in 1988, which included changes to the permanent disability system and compensability for stress-related injuries. The data show Massachusetts as having a comparably large drop in benefits in 1992. In 1991, Massachusetts - via Chapter 398 of the Acts of 1991 - made large changes to maximum benefit levels and durations, along with cost-of-living adjustments, compensability for stress claims, and waiting periods. The same can be said of what the data show for Montana in 1993. In 1992, the Montana government cut the period of wage replacement for certain injuries by 30%, from 500 weeks to 350. The government also made other changes to benefit formulae by specifying different allowances based on age, education, and experience.

Figure 7: Trends in Benefits Paid - Medians



Notes: This figure displays median workers' compensation benefits paid per \$1,000 of annual payroll within consistent private market states in panel (a) and new public option states in panel (b), across calendar time (year) and event time, respectively. A linear trend line is drawn through all points in panel (a) and drawn separately in panel (b) before and after formation (at time = 0). Vertical lines in panel (a) indicate public option creation years. Sources: CBP, SSB, and the National Academy of Social Insurance 1988-2020.

that there were large benefit drops in new public option states: a drop of 22% at the time of formation, from \$23 per \$1,000 of annual payroll in year $t = -1$ to \$18 per \$1,000 of annual payroll in year $t = 0$. Benefit levels themselves should be unaffected by a public option's entry into the market, both because any claim denial criteria apply to all insurers in a state and because benefit levels are legislated by the state, not personalized by insurers. Public options do not have any extra flexibility in which claims to (correctly) deny. Thus, these trends indicate that there may be room to make new public option states more comparable to their private market counterparts in terms of how much legislated benefit levels were changed.

5.2 Empirical Methodology

Given the legislated benefit movements, the most straightforward approach to better isolate the *unique* impact of the public option is to match each public option state to a host of consistently private-only states based on the percentage reduction in benefits paid amounts in the year a public option was formed. Then, by using a stacked event study estimator, I can average over more isolated, higher quality comparisons to provide an estimate of the

public option’s effects in the market. First, I will describe the general stacked event study methodology, then discuss the matching process and what it may be able to achieve.

5.2.1 Stacked Event Study Estimator

The methodology for this paper exploits a stacked event study design, employing difference-in-difference techniques, following work by [Deshpande and Li \(2019\)](#), [Cengiz et al. \(2019\)](#), and [Butters et al. \(2022\)](#). Typically, when treatment time varies, concerns mount for the traditional estimator ([Goodman-Bacon, 2021](#)); the stacked event study framework avoids problematic (“forbidden”) comparisons that come from allowing treatment time to vary without ensuring clean controls (that themselves do not move in and out of treatment). This setup is beneficial because it allows for pre-trend testing and the ability to study both contemporaneous effects, and how they develop over time. In the case of competition with a public option in insurance markets, the persistence of concentration effects, specifically, is critical to the discussion of the public option’s long-run behavior. There is also reason to expect heterogeneous effects over time, given each state’s experience before the public option is unique (with each state having its own insurance market).

The general framework for this type of setting is as follows:

$$Y_{s,t,k} = \alpha + \sum_{r=-4, r \neq -1}^{12} \beta_r (PublicOption_s * 1(RelTime_{t,k} = r)) + \gamma_{s,k} + \lambda_{t,k} + \epsilon_{s,t,k}, \quad (1)$$

where the outcome, at least in the main specifications, is either the HHI or loss ratio in state s , year t , and stack k . The event study framework appears with the β_r terms, where we can look back as far as four years prior to a change and as far forward as twelve years after a change (given Maine created a public option in 1993 but then made changes to it in 2007, with a buffer for anticipatory effects). The $PublicOption_s$ dummy variable indicates whether a state is a new public option or consistent private-only observation, which is interacted with the $RelTime_{t,k}$ indicators. These relative time indicators track the years before and after

public option creation (at $r = 0$), where t tracks the year and k defines the stack’s treatment year. The interpretation of the estimates represents the difference between the two types of states over time, relative to the year just before public option creation.

In the stacked event study framework I propose, only states that create a public option for the first time during the sample period and states that have never operated one during the study period are included in the main analysis sample. In Equation 1 above, there are six stacks: one for each new public option state and its matched comparators from among the consistently private market group. Ultimately, stacks are mini experiments and the stacked event study specification works across all of them to identify effects in relative time. The specification includes both state-stack and year-stack fixed effects. Clustering is done at the state-by-stack level, which is especially important given repeat observations do occur among the consistently private group.

Recent advancements in staggered timing econometrics suggests that not weighting stacked event study estimates with varying treatment group sizes across stacks means the target parameter (i.e., the average treatment effect on the treated or ATT) is not identified (Wing et al., 2024). In my context, each stack contains the same number of “treated” observations (i.e., new public option states), so a special weighting design is not necessary.

5.2.2 “Matching” on Legislated Benefit Cuts

As shown in earlier figures, pure levels of benefits are different across states. In the setting of trying to better isolate the public option’s unique effects, however, the goal is to match each public option state to a set of private-only states that best reflect the percentage change in benefits paid in the states that created a public option. This matching design will not be perfect, however, given that public option states were more often operating from larger benefit levels and so their percentage cuts were comparably larger. In this way, the matching exercise is done to provide some insight into how making states at least *morecomparable* may change the results relative to not matching. This will mean that reductions in benefits

may still have some explanatory power on any eventual result, given the larger changes in public option states, but that any change in results from an unmatched to a matched setting implies that benefit levels were confounding with the public option formation.

Table 1 below describes the matching that was done in this case whereby, for each new public option state, I document - for both the new public option state and its matched comparators - percentage decrease in benefits in the year of public option formation (with levels), along with statewide average loss ratios and payroll size in the year before public option formation.

The matching exercise identifies the states as they were matched; the basis for matching comes from the sixth column, which is the percentage change in benefits paid in the year of public option formation. With a small sample of states to start, similarity of states on other dimensions (like benefits levels, payroll, and loss ratios) is more difficult to achieve. As expected, new public option states tend to have both higher benefit levels (before and after) and higher loss ratios in the year before formation. The comparison of payroll is a bit more mixed, given that it varies quite widely within the matched comparator group in several instances.

Appendix Figure A.7 shows each public option state's benefits paid level over time, which - before matching - would be relative to the average of all consistent private market states. After matching, on a state-by-state basis, Appendix Figure A.8 provides the matched comparisons. Another way to understand both the power of matching in this example and the incompleteness of such a comparison in this context is to apply the stacked event study estimates - matched and unmatched - with benefits paid as the outcome. Figure 8 provides those results.

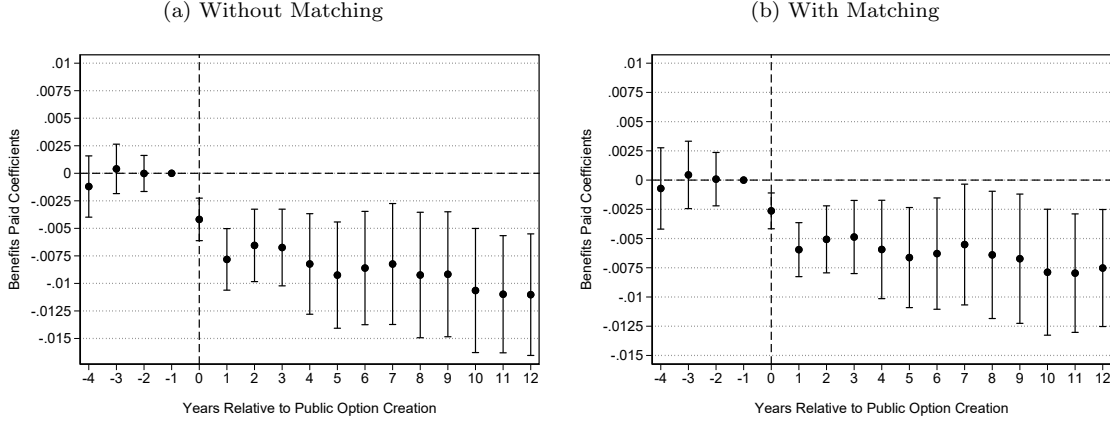
As is clear from these event study results, matching closes the gap in benefits paid effects that were experienced differentially in new public option states relevant to states that were consistently private-only. Most of the point estimates in the matched results imply a 25-30% reduction in magnitude from the unmatched size, indicative of the role matching

Table 1: Matching Exercise

Public Option State	Formation Year	Matched Comparators	Benefits Paid Year Before	Benefits Paid Year Of	% Change in Benefits Paid	Statewide Average Loss Ratios, Year Before	Statewide Total Payroll, Year Before
Louisiana	1992	(5): Arkansas, Florida, Indiana, Massachusetts, Wisconsin	T: \$21.0 C: \$16.4	T: \$18.3 C: \$14.6	T: -12.9% C: -10.6%	T: 0.96 C: 0.90	T: \$26,711,019 C: \$52,232,873
Texas	1992	(4): Arkansas, Florida, Massachusetts, Wisconsin	T: \$24.0 C: \$18.4	T: \$18.4 C: \$15.9	T: -23.3% C: -11.3%	T: 0.80 C: 0.96	T: \$136,202,165 C: \$53,749,127
Maine	1993	(2): Massachusetts, Mississippi	T: \$50.1 C: \$17.3	T: \$38.3 C: \$14.4	T: -23.6% C: -21.7%	T: 1.10 C: 0.72	T: \$8,888,975 C: \$42,754,122
Missouri	1995	(7): Connecticut, Florida, Illinois, Indiana, Kansas, New Hampshire, Virginia	T: \$15.7 C: \$14.5	T: \$13.7 C: \$12.6	T: -12.7% C: -12.6%	T: 0.68 C: 0.68	T: \$50,161,688 C: \$62,375,379
Kentucky	1995	(5): Georgia, Massachusetts, Nebraska, North Carolina, Tennessee	T: \$20.6 C: \$10.7	T: \$16.6 C: \$8.7	T: -19.4% C: -19.2%	T: 1.01 C: 0.60	T: \$28,324,513 C: \$54,061,563
Hawaii	1997	(2): South Dakota, Tennessee	T: \$27.4 C: \$12.8	T: \$22.8 C: \$10.8	T: -16.7% C: -15.8%	T: 0.70 C: 0.63	T: \$10,954,149 C: \$30,259,467

Notes: Author's documentation of matching exercise with author's own calculations where necessary. "T" refers to the value in the public option state; "C" refers to the average value among the matched comparators. Benefits paid columns are per \$1,000 of annual payroll. Payroll column is in thousands of US dollars. Source: CBP, SSB, National Academy of Social Insurance, and NAIC 1991-1998.

Figure 8: Stacked Event Study Results - Benefits Paid



Notes: This figure displays point estimates from estimating Equation 1 using annual benefits paid per \$1 of payroll as the outcome for two specifications: unmatched in panel (a) and matched in panel (b). 90% confidence intervals are used. Sources: CBP, SSB, and the National Academy of Social Insurance 1988-2020.

plays in making states more similar. Yet, the matching is not perfect in that panel (b) still shows statistically significant benefits paid changes in public option states over their matched comparators. Thus, while an imperfect match will mean that loss ratio (and potentially for HHI, too) results may continue to be explained by larger reductions in benefits levels for public option states, matching does substantially close the gap between the sets of states, so any meaningful difference in results from matching are quantifying the role benefits legislation plays and identifying something closer to the public option’s *unique* effect than if there were no matching.

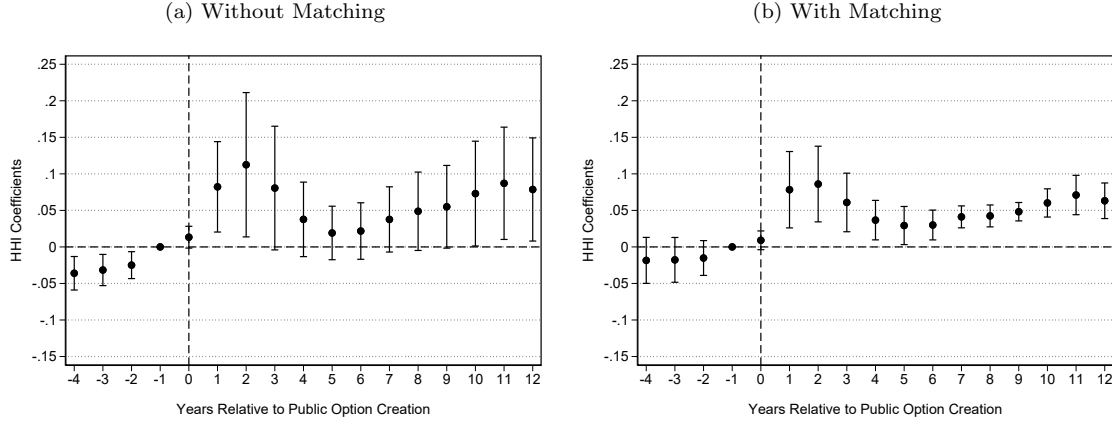
6 Results

Below, I provide stacked event study results for the two main outcomes of the paper: HHI and loss ratios. Using Equation 1, I provide the main results in both unmatched and matched form.²¹ In each of these event study depictions, the estimates are provided in event time, where public option formation happened at $time = 0$, and the effects presented represent the difference between new public option states and consistently private-only states relative to the baseline difference in the year just before formation. First, Figure 9 presents the HHI

²¹In the unmatched results, weights of each stack - of which there are four (1992, 1993, 1995, and 1997) is based on the percentage of new public option states in each stack, as per Wing et al. (2024)

results.

Figure 9: Stacked Event Study Results - HHI



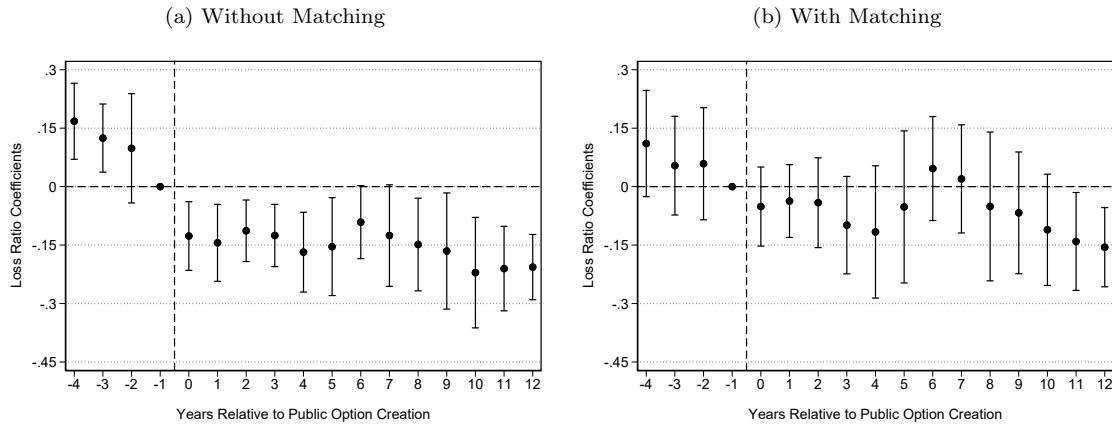
Notes: This figure displays point estimates from estimating Equation 1 using statewide HHI as the outcome for two specifications: unmatched in panel (a) and matched in panel (b). 90% confidence intervals are used. Sources: NAIC 1988-2020.

Both panels present very similar results, indicating that the main driver of the HHI results is the public option itself, not simultaneous benefit changes. While the coefficient magnitudes do not change much from panel (a) to panel (b), the standard errors do. This is indicative of the power of matching in minimizing some of the large variation in the concentration experiences among the consistently private-only states. Matching also helps with comparability; this is especially true in the pre-period where the trends in advance of formation are no longer statistically significant after matching. In this case, it is clear that the public option is responsible for significantly increasing the concentration of the workers' compensation insurance markets in the states in which they were formed. The changes are not only statistically relevant; with the baseline HHI gap between the two states at about 0.05, these effects indicate a doubling of the HHI that is persistent over the decade that follows public option formation. This is above and beyond the already elevated concentration levels that may have helped inspire public option formation in the first place.

Figure 10 presents the results for loss ratios. When looking just at the unmatched results in panel (a), the effects seem similar to what would have been expected from the more

descriptive evidence presented earlier. Namely, it appears that new public option states caused improved insurer loss ratio experiences in both economically and statistically relevant ways. Pre-trends, however, are statistically different from zero, which speaks to the concern about other changes around this time that may be driving the results.

Figure 10: Stacked Event Study Results - Loss Ratios



Notes: This figure displays point estimates from estimating Equation 1 using statewide average loss ratios as the outcome for two specifications: unmatched in panel (a) and matched in panel (b). 90% confidence intervals are used. Sources: NAIC 1988-2020.

Controlling for major legislated benefit level changes at the same time by matching on the percentage benefit change in the year of formation, panel (b) largely indicates a disappearance of the improvements in loss ratios. And, importantly, it is not all due to the decreased sample size. While confidence intervals tend to be larger, every coefficient has shifted toward zero. As shown above, the matching process reduced benefit reduction disparities between the two groups by about 30% in most post-formation years. Only using the coefficients from the loss ratio estimates suggests a cut in magnitude of more than 50% in most periods. Thus, while benefit reduction disparities were not reduced to zero, moving toward a world where the public option is more isolated as a policy lever (separate from legislated benefits) shows that the public option had little distinct effect on average statewide loss ratios, even when the effect on concentration is clear, robust, and (economically and statistically) significantly positive.

7 Mechanisms & Measuring Price

There are several potential explanations for the results presented above. The HHI results, for example, could be explained in two ways: a) the public option absorbs “bad risks” as the residual market mechanism (increasing its market share mechanically) and allows the private market to compete in a healthy way or, in a more cynical view, b) the public option erodes competition and eats up non-residual market share (forcing private insurer exits or acceptance of smaller market shares). The results above made it clear that legislated benefit level changes are not a driver of the HHI effects I have documented.

As shown above, a large part of the explanation for the loss ratio results that could have initially been attributed to the public option was actually coming from changes in differences in legislated benefit level cuts. Yet, the loss ratio coefficients were still fairly consistently negative and would be economically relevant, even if not statistically different from zero (though they are in later years). This can be explained in several ways: a) the additional disparity in legislated benefit cuts (due to imperfect matching) sufficiently outweighs changes in prices; b) shifting bad risks to the public option (raising its loss ratios) reduces private insurer loss ratios; c) the public option is able to operate at an advantage that, combining increased market share and higher prices, shifts statewide underwriting profitability with the performance of the public option; and/or d) public option states saw differential shifts in industry mix, particularly away from unsafe industries.

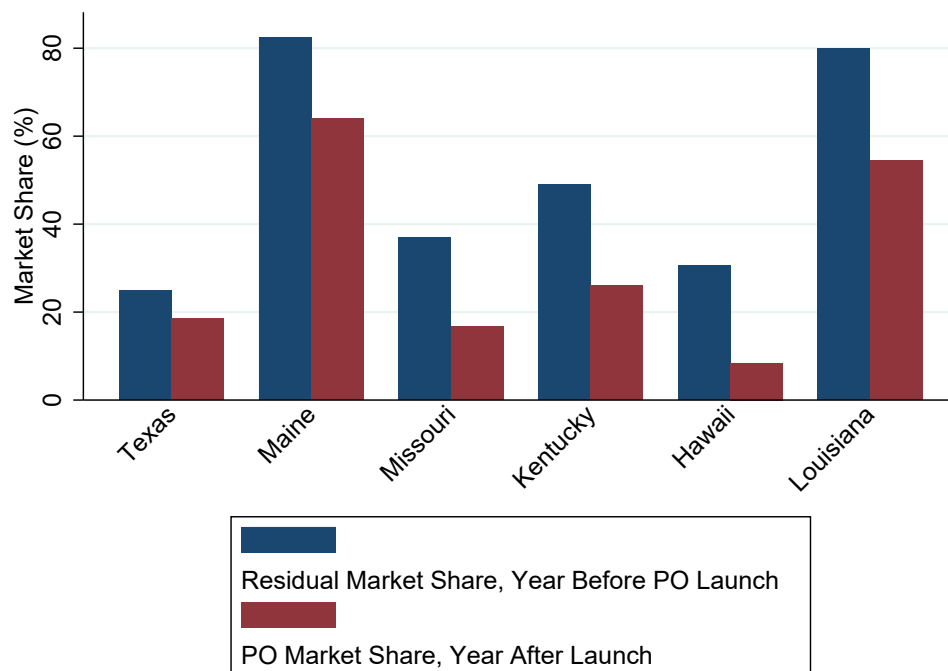
One item on this list that cannot be added, however, is that public options join the market and either deny claims at higher rates or allow the private insurers to do so. In workers’ compensation, there is no way for the insurer to provide less generous benefits than what the state prescribes and the legal foundation for being paid on a workers’ compensation claim as an injured worker remained unchanged. If the compensability of certain types of claims was changed legally, it would apply to all insurers in a state. Thus, when I refer to legislated benefit reductions in this context, it really means more about controlled healthcare options, less income replacement, and/or shorter benefit durations. In what follows, I provide

evidence for or against each rationale explained above.

7.1 Ceding “Bad Risks” to the Public Option

The public option accepting bad risks could explain both the public option’s strong effects on HHI and persistent loss ratio patterns. One way to understand if the public option is simply accepting the “bad risks” (increasing concentration) while allowing the private market to perform better (lowering statewide average loss ratios) is to look at the residual market share in new public option states before public option formation and compare that to the eventual market share of the public option. Figure 11 does exactly that, and Appendix Figure A.9 does something similar for each set of matched comparator private-only states, showing residual market shares in the year of public option launch and the three years that follow.

Figure 11: Residual Market Share Before vs. Public Option Market Share After



Notes: This figure displays residual market shares in the year before the public option (PO) launched, compared to the PO’s market share in the year after launch. Sources: NAIC 1992-1998, [Thomason et al. \(2001\)](#), [Buhro \(2011\)](#), and [NCCI \(2000\)](#).

As is clear from the figure, and the large drops in residual market shares that occurred

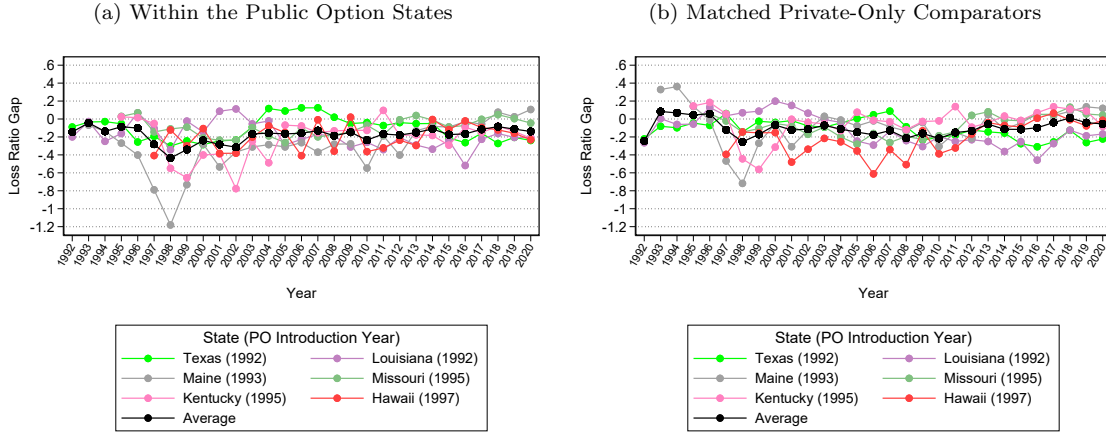
in the matched private market states, public option market shares do not reflect a simple shifting of “bad” risks from the residual market to the public option. In fact, it appears that - maybe due to benefit reductions for all insurers in each state - the private market became willing to write previously residual risks as standard market contracts (which must be the case in the matched private market states, where residual market went from north of 40% in the matched year of formation to less than 15% within three years in most cases). Certainly, some of the public option’s market share could be made of up residual risks, which is why it is the focus of the next set of evidence to take up the performance of the public option compared to the private market.

Another way to get at this point is to compare the loss ratios of the public option to the loss ratios of its matched private competitors. If this explanation is true, the public option may not be performing better than the rest of the market (if price frictions exist), given they do not have the ability to reject more claims or decrease benefits for injured workers on their own. Figure 12 explores this dynamic by asking about the gap between the loss ratio of the public option and the average statewide loss ratio of the private market (i.e., $LossRatio_{PublicOption} - LossRatio_{Private}$), where positive values reflect higher loss ratios (worse underwriting performance) for public options and negative values reflect stronger underwriting performance for the public option than the private market average.²² This is done in two cases - compared to the average statewide loss ratio in the public option’s state (with the public option removed) in panel (a) and compared to the average statewide loss ratio in matched private-only states in panel (b).

In panel (a), public option loss ratios are consistently below that of their private market competitors within their home states. This is an important comparison because it is within a legislated benefit regime, wherein each insurer faces the same legislated benefit levels (and was exposed to the same changes over time). While there is some variability in public option loss ratios year-over-year (which is common at the firm-level in all lines of insurance), the

²²Appendix Figure A.10 depicts the baseline average loss ratios for each public option over time, for reference.

Figure 12: Loss Ratio Gaps - Public Options vs. the Private Market



Notes: This figure displays the gaps between the loss ratios of the public option and the private market over time. The private market is defined as all private insurers in the public option's state in panel (a), and the matched group of private-only comparators in panel (b). Sources: NAIC 1992-2020.

average gap is negative in every year. In many years, the magnitudes are large - and not just in Maine where there may be an outlier in public option financial reporting in 1998. Even a 20% gap in the loss ratio is quite large economically. When compared to the private market average in the matched comparator states via panel (b), the picture is largely the same but with a few additional positive gap patterns. This comparison is weaker than panel (a), given that the matched comparator states exist under different legislated benefit regimes over time.

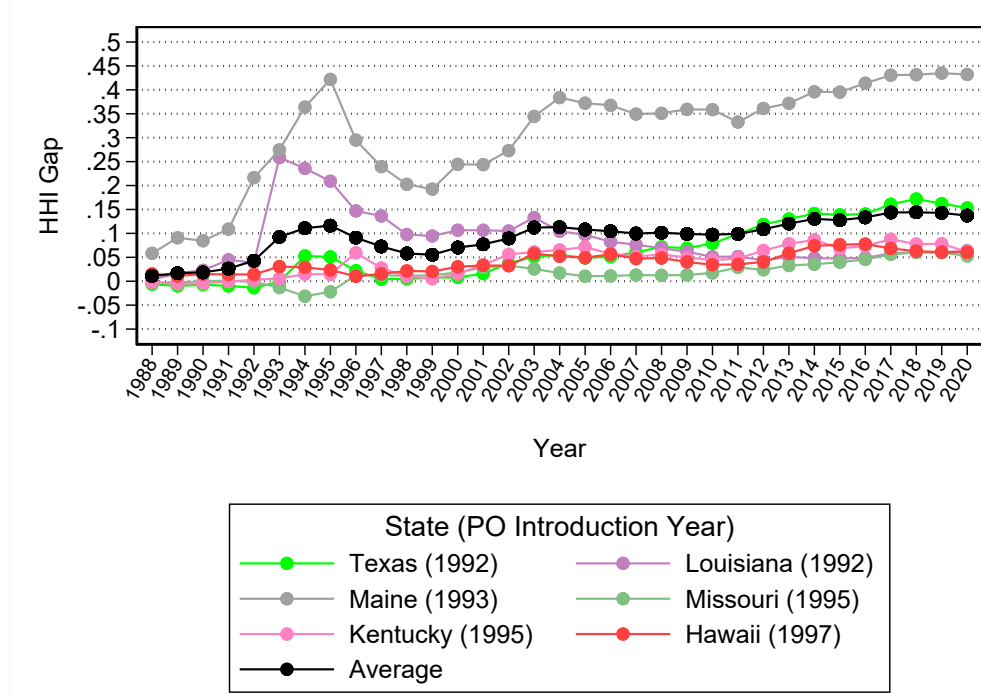
The evidence from Figure 12 suggests that enhanced underwriting profitability and increased concentration may not come from the public option purely accepting worse risks. In fact, public options appear to have better loss ratio experiences than their private market counterparts in a robust way, consistent across states and over time. Additional evidence supports this idea. Appendix Figure A.11 duplicates the main, matched event study results for loss ratios, with (like Figure 10) and without the public options included). Results indicate that after the first few years, not including the public option further nullifies the loss ratio effect (even making the coefficient statistically significant and positive in one year), further verifying that the public option performance is as good as, and likely better than,

the private market.

Beyond not simply taking all of the residual market share and outperforming the private market in terms of loss ratios, one final way to get at the point about whether public options take all bad risks is to look at long-term trends in concentration measures. As documented earlier in Figure 3, public option market share has continued at high levels even through the nearer term. Appendix Figure A.12 displays the national residual market share of all states which NCCI services (26) over a recent set of years (2012-2020). In each year, residual market share never exceeds 10%. Public option market share, on the other hand, exceeds 20% in all six of my main sample's new public option states during that time span. Appendix Figure A.13 displays HHI over the entire panel - 1988-2020 - and indicates that the concentration levels of public option states remains high even into the near-present. To get at this point even more clearly, Figure 13 shows the gap in HHI between new public option states and their matched private-only comparators (i.e., $HHI_{PublicOptionStates} - HHI_{PrivateMarketMatch}$). As is clear, public option states are consistently more concentrated than their matched private comparators (and, economically, the gaps are large - the average HHI in each matched stack never exceeds 0.04).

Taken together, the evidence in this section suggests that the public option simply accepting the worst risks (from the previously-operating residual markets) is not a compelling explanation for the HHI effects or loss ratio patterns. Public options did not appear to take all of the residual market share when they were first formed, and legislated benefit cuts may have made residual risks more appetizing to the private market. Beyond the immediate shocks, public options have continuously outperformed the private market (within their home states and relative to matched private-only comparator states) while retaining large relevance in the market, far more than can be attributed to residual markets using current national averages.

Figure 13: HHI Gap - New Public Option States vs. Matched Private Market States



Notes: This figure displays the gap in HHI between new public option states and matched private market comparators over time. Sources: NAIC 1988-2020.

7.2 Entries and Exits

Another way to tease out what dynamics are behind the increased concentration of the marketplace is to ask about insurer movement in and out of the market. The public option can gain market share by forcing exits or by actively competing, winning, and gaining market share while the private market accepts a smaller role. In Appendix Figure A.14, I consider the impact of the public option on the net entry ratio: $(N_{Entries} - N_{Exits})/N_{Insurers}$. These results, while indicative of perplexing pre-trends, indicate that entries increased after the public option, more than enough to offset the exit experience. While unlikely to be causal, these results suggest that the public option is not gaining market share by forcing exits. One explanation for these results could stem from the fact that workers' compensation insurers may sell other types of insurance to their commercial clients, allowing them to bundle coverage, so that there is a benefit of remaining in the market even if at a reduced

presence. In sum total, the strong concentration results I document tend to be driven by active competition from the public option.

7.3 Changes in Statutory Benefits and Price Dynamics

While there is little empirical support for the idea that loss ratios decrease because of siphoning bad risks to the public option, we have seen that statutorily-mandated benefits levels were decreased in many states around the same time as public options were created. There is little reason to expect that data on benefits paid should show a reduction through any other channel than via statute. Thus, one potential explanation for the persistent negative coefficients with respect to the loss ratio (even if they are not always statistically significant) - which may also work in conjunction with the public option consistently outperforming the private market - is that, even after matching, new public option states experienced out-sized benefit reductions relative to price changes.

To test this, I need to construct a price measure. Price is not straightforward to measure because premiums, what one may think of as price, are a function of both price and quantity. And in many lines of insurance, quantity is an unknown component of premiums because coverage is not always uniform and the number of “units sold” (i.e., clients) are often unknown. In financial data, which is commonly used in insurance economics via the NAIC (as I use in this paper), we do not know how many cars, homes, people, business, etc. are covered by the reported premium amounts. Workers’ compensation, however, presents a unique ability to better estimate price because we largely know the quantity: we know how much payroll exists in each state in each year, and workers’ compensation is required in virtually every state (meaning almost all payroll is covered) and its coverage is standardized (within a state, medical benefits and lost income replacement are consistent across employers). One caveat to all payroll being covered is that some payroll (potentially significant amounts, varying by state) are covered by self-insured employers. Yet another advantage of workers’ compensation is the availability of data on how much self-insured employers pay in benefits in each year and state.

Thus, making use of all the available data, I suggest the following price measure:

$$\text{Price}_{s,t} = \frac{DPW_{s,t}}{\text{InsuredPayroll}_{s,t}}, \quad (2)$$

where the price of workers' compensation insurance in state s in year t is equal to the total direct premiums written²³ per dollar of insured payroll, which is defined as:

$$\text{InsuredPayroll}_{s,t} = \text{TotalPayroll}_{s,t} * (1 - \frac{\text{SelfInsuredBenefitsPaid}_{s,t}}{\text{TotalBenefitsPaid}_{s,t}}).^{24} \quad (3)$$

Before moving to the traditional event study estimator to compare the price response and the benefits paid response, it is necessary to provide trend analogs in price. Appendix Figures A.15 and A.16 do so for median and mean prices per \$1 of payroll, respectively. The trends indicate broad price reductions in both groups of states, with a clear drop after the public option enters the market that returns new public option state unit prices to the private market comparator average. These comparable price drops in both locations largely match the trends in benefits paid presented earlier. These trend figures do not, however, tell us clearly about any differential changes between the two groups of states that exist after matching.

Figure 14 presents formal (matched) event study results for price and benefits paid (a repeat from Figure 8), both per \$1 of annual payroll and graphed on the same set of axes.²⁵ Panel (a) indicates that prices were reduced by about \$0.005 after the public option entered the market with some variation over time. Panel (b) suggests that the reduction in benefits paid ranges from \$0.005 in the first year post-creation to \$0.0075 for the rest of the period.

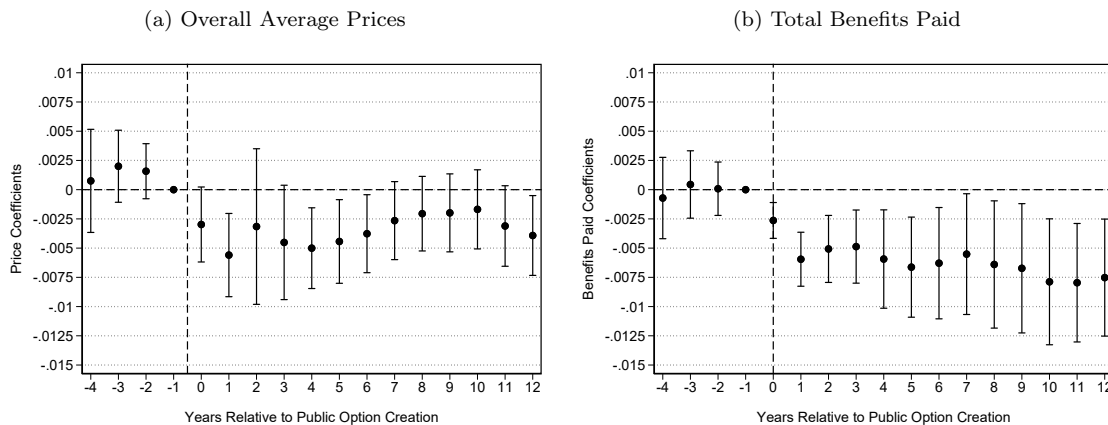
²³All elements of this equation are first moved to be in 2020 US dollars using the consumer price index.

²⁴It is possible to modify this setup in several ways. For instance, one could specify insured payroll as privately-insured or PO-insured payroll because insured benefits paid are split by privately-insured and those insured by the public option. One could also adjust the direct premiums written numerator by subtracting dividends paid from the insurer back to the employer, creating an adjusted price measure.

²⁵Appendix Figure A.17 presents matched event study results for the self-insured percentage of benefits paid, an adjusted price measure (factoring dividends out of direct premiums written), and dividends paid per \$1,000 of annual payroll. Though imprecise, there is a general trend toward more self-insured benefits paid in new public option states over time. Prices adjusted for dividends look broadly the same as unadjusted prices, and there is little dividend difference over time.

When taken together, these results indicate that the negative loss ratio patterns persistent after matching may be driven by larger reductions in benefits paid to injured workers than reductions in prices paid by employers in new public option states than their matched private comparators. This is true in the short and long term.

Figure 14: Stacked Event Study Results - Price and Benefits Paid



Notes: This figure displays point estimates from estimating Equation 1 with two outcomes: price in panels (a) and annual benefits paid in panel (b). All outcomes are per \$1 of annual payroll. 90% confidence intervals are used. Sources: NAIC, CBP, SSB, and the National Academy of Social Insurance 1988-2020.

7.4 Changes in Industry Mix

If new public option states saw a differential change in their industry mix away from dangerous work, this could also explain sudden loss ratio improvements. Appendix Figure A.18 studies this dynamic and tests for changes in the percentage of a state's payroll coming from six key industry groups: mining, manufacturing, agriculture/fisheries/forestry, wholesale trade, retail trade, and construction. I use only these six industries because the industries are largely consistent across the CBP data series switching from Standard Industrial Classification (SIC) to North American Industry Classification System (NAICS) codes in 1998. These six categories make up between 40-60% of total payroll in a given state.²⁶ The results here, while noisy in some industries, largely indicate that industry mix is unlikely to explain the results. While there were some small decreases in mining, there were much larger

²⁶Appendix Figure A.19 shows the portion of total payroll captured, including the notable drop in 1998 as the NAICS codes are more granular than SIC codes.

increases in manufacturing and some increases in construction. There was little change in wholesale and retail trade, and in agriculture, fisheries, and forestry.

8 Role of the Public Option

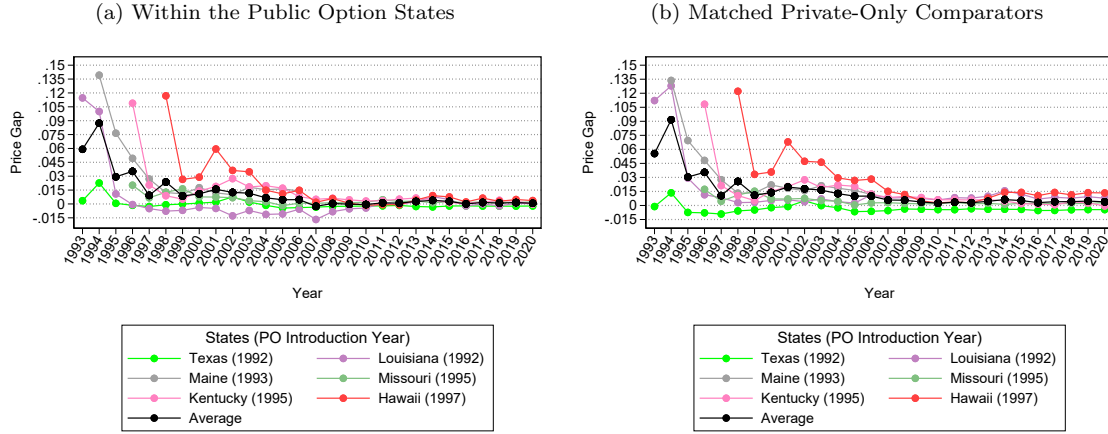
To this point, I have documented strong concentration effects that come - distinctly - from the formation of a true, competitive public option in workers' compensation markets. Public option formation leads to substantially more concentrated markets, not at the hands of other private insurers but the public option itself. At the same time, the public option has little distinct effect on statewide average loss ratios. The public option does, however, perform better than their private market competitors in most states and in most years since creation. Analysis of mechanisms suggests that public options don't only take on "bad risks," but that they compete and win market share. It appears statewide average loss ratio improvements mostly come from legislated benefit levels affecting all insurers, which were much larger than price changes in response.

This begs the question: what is the role of the public option in these marketplaces? It may have exacerbated competition-related market failures and had little role in affecting performance-driven market failures. Yet, these new public options continue to exist and take up large roles in their respective markets. I proffer two suggestions, and then provide evidence in support. Both suggestions are deserving of future research and continued study.

One role a public option could play is to either keep prices low or encourage efficient market segmentation. A natural step is to ask about the prices specific to public options relative to the rest of the market. Using the price measure I suggested in the prior section, Appendix Figure A.20 documents public option prices over time, outside of the first year of operation (which was characterized by overly high prices, as can be seen trickling into the next several years). Clearly, prices have come down substantially the longer the public options have been in the market. To better assess the difference in prices, I ask about the price gap in Figure 15 (i.e., $Price_{PublicOption} - Price_{Private}$).

The price gaps in this main figure indicate that, generally, public options tend to charge

Figure 15: Price Gaps - Public Options vs. the Private Market



Notes: This figure displays the gaps between the public option price and the private market price over time. The private market is defined as all private insurers in the public option’s state in panel (a), and the average private price from the matched group of private-only comparators in panel (b). Sources: NAIC, CBP, SSB, and the National Academy of Social Insurance 1988-2020.

roughly the same or even higher prices than their private market counterparts, either within their home state in panel (a) or with their matched private-only counterparts in panel (b). This is clear through about 2008, when it becomes more challenging to clearly see the gaps. Appendix Figure A.21 zooms in on the 2008-2020 period to show that this pattern in positive gaps (higher prices for the public option) persist, though they are smaller in magnitude.

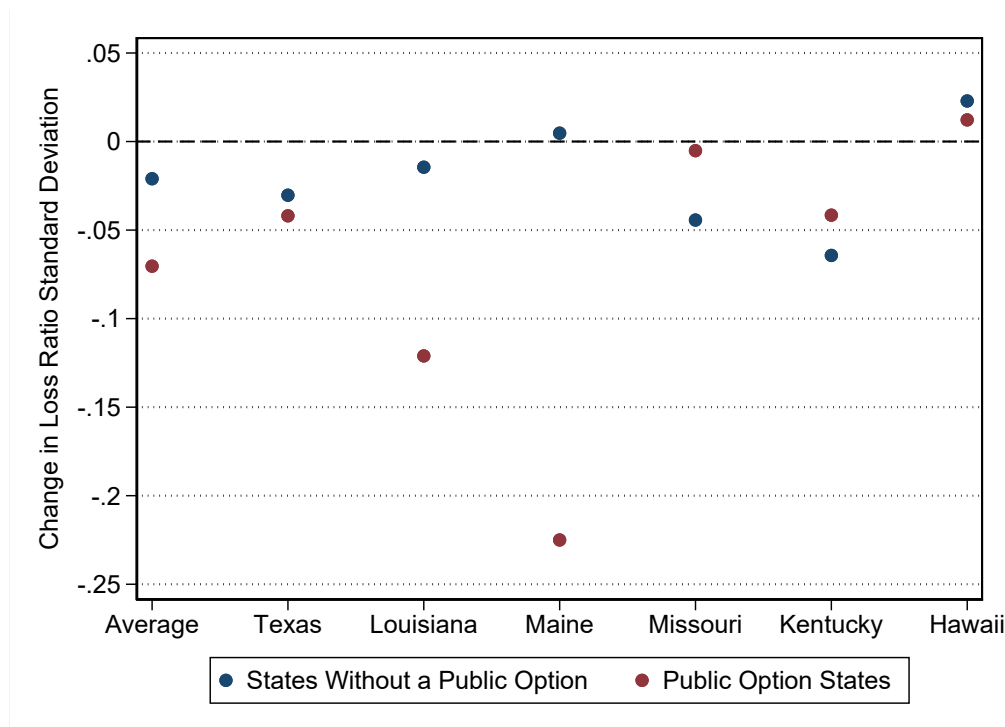
These price gap results suggest that public options are operating at such large market shares, which have largely grown over time and remained high, not by charging systematically lower prices than their private competitors. In essence, public options are competing, winning business, and performing quite well with rates at least as high as the private market, if not usually higher. This does not mean that the public option is not serving the role of keeping prices down, however, given that workers’ compensation is highly experience rated. Thus, they may charge higher prices because they tend to underwrite higher risk industries (even if these are not “residual” risks). This could be seen as evidence of market segmentation in public option states, which is a promising endeavor for future research. Future research would also be wise to consider the ways in which public options can gain market share at higher prices, which could be related to loss mitigation expertise.

A separate role the public option could play in the market is to stabilize the loss ratio experience of private market insurers. This is a different question than asking about the loss ratio experience of the state in terms of a statewide loss ratio. Rather, this role is more about firm-level loss ratios, which are commonly quite noisy. To get more at this point, I look at the variation (via the standard deviation) in firm-level (not including public options) loss ratios in the three years before the public option is created and compare that to the variation of firm-level loss ratios in the year of creation and the following two years (i.e., three years on either side of formation).

Figure 16 presents the result using insurers who operated for the entire three year period before or the entire period after formation (i.e., not allowing entering/exiting insurers to affect the results). Appendix Figure A.22 uses all firms, regardless of entry/exit behavior and the results are broadly consistent, though the changes are smaller, likely given the increased variability for new or exiting firms. In both cases, firm-level loss ratios outside of the 10th and 90th percentiles are dropped (given there are extreme outliers in the data). In the graphs, the blue dot within each grouping represents the change in loss ratio standard deviation (i.e., $SD_{After} - SD_{Before}$) in matched private market comparator states (i.e., matched to the new public option state given on the x-axis); the red dots show the difference among private insurers in new public option states (given on the x-axis). Arguably, the blue dots provide an estimate of the improved stability (a negative change would be an improvement in this depiction) that comes from benefit reductions and the red dots would provide a more focused benefit for the public option. The average of the six gaps among each group is graphed on the far left.

The results from this exercise suggest, on average, the public option may have played a role in stabilizing the loss ratio experience of private insurers within their home state. This role, however, is not robust across all six states. In two states (Missouri and Kentucky), the improvement in firm-level loss ratios was larger in matched states without a public option than among private insurers in the public option states. In other states, the public option

Figure 16: Firm-Level Stability in Loss Ratios



Notes: This figure displays the change in the standard deviation of firm-level loss ratios between the three years before public option formation and then the three years after (including the year of formation) among private insurers in matched private-only states (blue dots) and states with a new public option (red dots). States are grouped by their reference public option (x-axis), along with the average of the gaps across all six categories on the far left. This graph drops firm-level loss ratios outside of the 10/90 percentiles, and only firms operating in all three years before or all three years after are used. Sources: NAIC 1989-1999.

may be related to more stability (or less instability, in the case of Hawaii).

There could be several explanations for these patterns, many of which are beyond the scope of this paper. Differential benefits paid experiences after matching, for example, could play a role if there were magnitude differences that persisted after the match. Figure A.8 shows some of those differences, but they do not appear to clearly tell a story with the stability estimates. Another possibility is additional market changes, like shifts to open competition (rather than administered pricing, which allowed very little pre-contract price competition) around the same time as public option formation. Using tracking from [Thomason et al. \(2001\)](#), this also appears to explain little of the pattern.²⁷ Other changes may also

²⁷Maine, Texas, and Missouri moved to open competition in the year around public option formation. Louisiana, Hawaii, and Kentucky had been operating in an open competition setting for at least four years before formation. While matched comparator states could also have changed regimes, and sometimes did,

explain these shocks to stability. As such, this question about the stabilizing role of the public option in workers' compensation insurance markets is one worthy of future researcher attention.

9 Conclusion

Public options are often discussed as policy tools in US insurance markets, especially in the health context. Despite the attention, however, there exists little empirical evidence of the impact on markets when public options are formed. In this paper, I provide (to my knowledge) the first empirical evidence as to the effect of a standalone, competitive public option in US insurance markets by utilizing the workers' compensation arena.

In workers' compensation insurance markets, several states created public options between 1992 and 1997. I leverage these changes to study the development of competition and insurer loss ratios in new public option states relative to states that have a market made up of only private carriers. I document that while new public option states experienced rising concentration leading to the policy change, concentration significantly increased after the public option entered the market. While insurer underwriting profitability appeared positively affected by the public option, matching on simultaneous legislated benefit changes provides little room for a standalone effect of a public option on average loss ratios.

In testing various mechanisms, I provide evidence that increasing concentration is driven not by the guaranteed issue nature of public options but via active competition and increased market share, much more than can be explained by their residual market role. Insurers are not exiting at higher rates, though, and tend to continue entering the concentrated market. Any improvements in performance that exist (even if not statistically significant) seem to be driven by the fact that even matching is imperfect here: new public option states made larger benefit cuts on average than private-only states, so average performance tended to improve differentially because price drops did not match the severity of benefit cuts.

there appears to be little role there. For example, Louisiana and Texas have the same four states as matched comparators, except that Louisiana has a fifth matched state, Indiana (which did not make a pricing change around formation time).

This work has important policy ramifications in workers' compensation insurance debates. Two states have recently changed their public option rules, with Arizona completely privatizing its public option just over a decade ago and Missouri having just passed legislation to do so for its public option by 2025, for example. In addition, by laying out two rationales for continued public option market involvement (despite large market shares and high levels of profitability), this work outlines important areas for future work related to a public option's ability to manage market prices and stabilize private insurer performance.

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Chapter I Appendix Tables and Figures

Intended for online-only publication.

Table A.1: Comparing Non-Negative DPW Firm-Years to Negative DPW Firm-Years

	(1) Non-Negative	(2) Negative
Member of Group	0.92 (0.27)	0.96 (0.20)
Stock Insurer	0.86 (0.35)	0.92 (0.27)
Mutual Insurer	0.11 (0.31)	0.07 (0.25)
Magnitude of Negative (\$1000s)	0.00 (0.00)	-266.99 (1450.54)
Direct Premiums Earned (\$1000s)	4792.06 (23542.25)	-112.99 (1640.76)
Dividends (\$1000s)	143.46 (2712.35)	13.02 (335.86)
Direct Losses Paid (\$1000s)	2659.60 (14003.95)	981.28 (6214.04)
LAEs Paid (\$1000s)	268.10 (1564.76)	108.65 (773.28)
Direct Premiums Written (\$1000s)	4848.97 (23711.46)	-266.99 (1450.54)
<i>N</i>	437905	27578

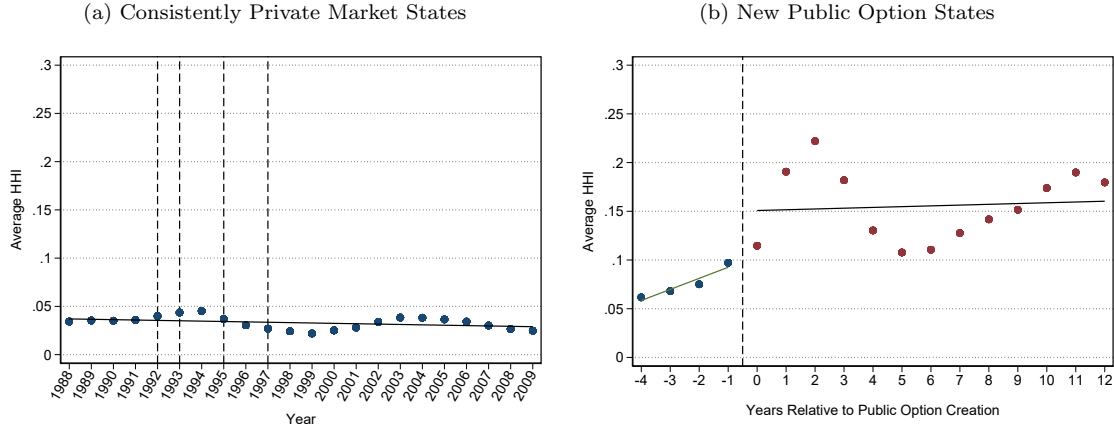
Notes: Depicted are sample averages across variables (rows) and negative DPW status (columns). Standard deviations are in parentheses. Dollar amounts in 2020 US dollars. Source: NAIC 1988-2020.

Table A.2: Comparing (N)ever Negative Firms

	(1) Non-Negative	(2) Negative
Member of Group	0.89 (0.31)	0.96 (0.21)
Stock Insurer	0.82 (0.38)	0.90 (0.30)
Mutual Insurer	0.13 (0.34)	0.08 (0.27)
Magnitude of Negative (\$1000s)	0.00 (0.00)	-29.37 (488.32)
Direct Premiums Earned (\$1000s)	6675.36 (30669.98)	2634.84 (12812.70)
Dividends (\$1000s)	213.66 (3700.53)	69.63 (1119.49)
Direct Losses Paid (\$1000s)	3612.94 (18586.72)	1668.64 (7260.79)
LAEs Paid (\$1000s)	351.40 (1979.47)	179.92 (1001.44)
Direct Premiums Written (\$1000s)	6764.11 (30860.66)	2639.22 (12959.98)
Avg. of Neg. DPW Firm-State-Years when Negative	. (.)	-262.91 (1021.70)
Avg. of Neg. DPW Firm-State-Years when Positive	. (.)	3063.39 (9657.79)
<i>N</i>	235070	230413

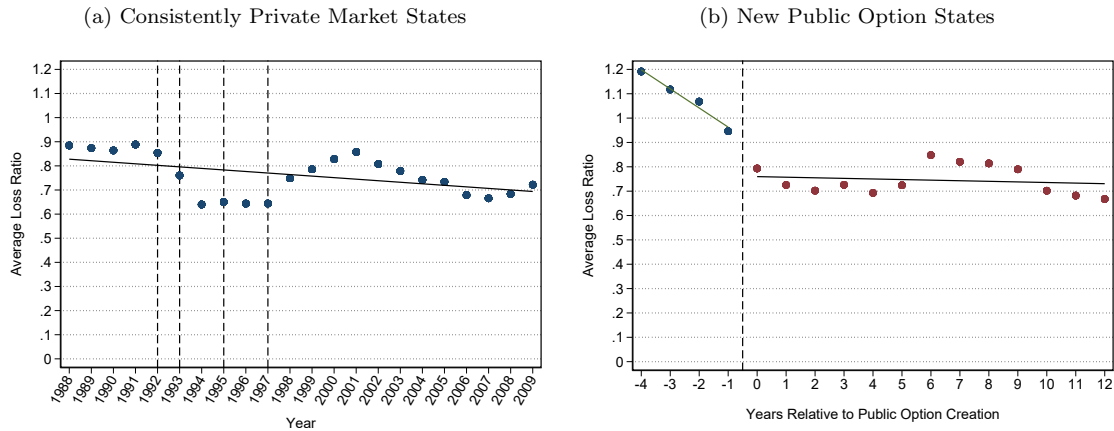
Notes: Depicted are sample averages across variables (rows) and (n)ever negative DPW status (columns). Standard deviations are in parentheses. Dollar amounts in 2020 US dollars. Source: NAIC 1988-2020.

Figure A.1: Trends in Concentration (HHI) - Means



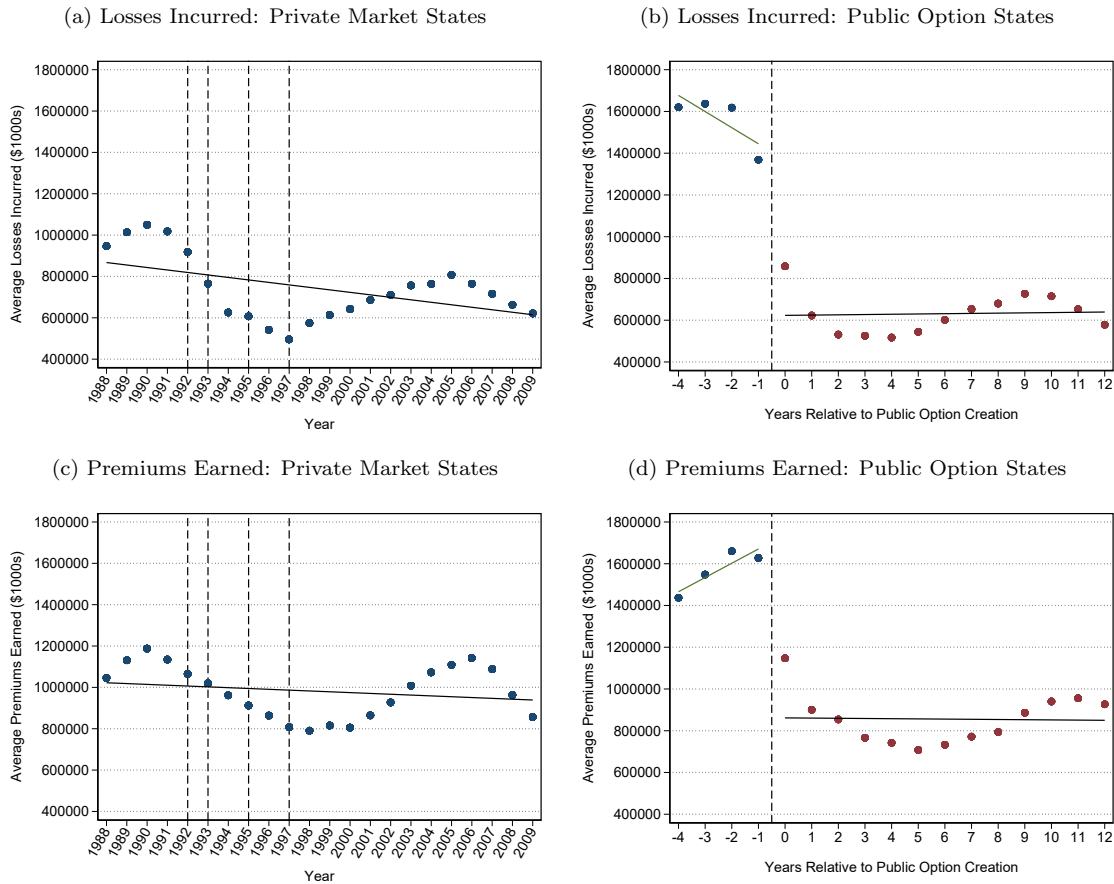
Notes: This figure displays average HHIs across states in consistently private market states in panel (a) and new public option states (b), across calendar time (year) and event time, respectively. A linear trend line is drawn through all points in panel (a) and drawn separately in panel (b) before and after public option formation (at time = 0). Vertical lines in panel (a) indicate public option creation years in public option states. Sources: NAIC 1988-2020.

Figure A.2: Trends in Loss Ratios - Means



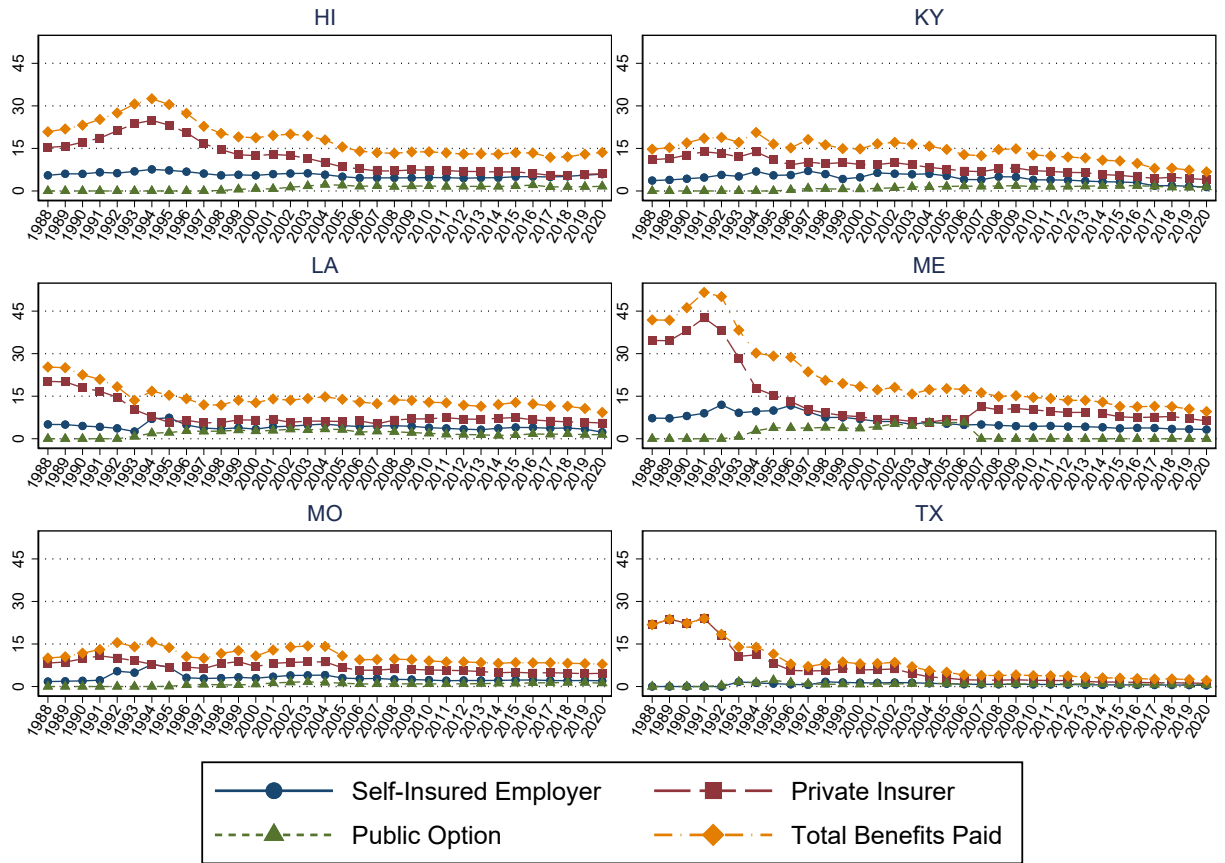
Notes: This figure displays mean loss ratios for consistently private market states in panel (a) and new public options states in panel (b), across calendar time (year) and event time, respectively. A linear trend line is drawn through all points in panel (a) and drawn separately in panel (b) before and after formation (at time = 0). Vertical lines in the left-side graphs depict public option creation years. Sources: NAIC 1988-2020.

Figure A.3: Trends in Loss Ratio Elements - Means



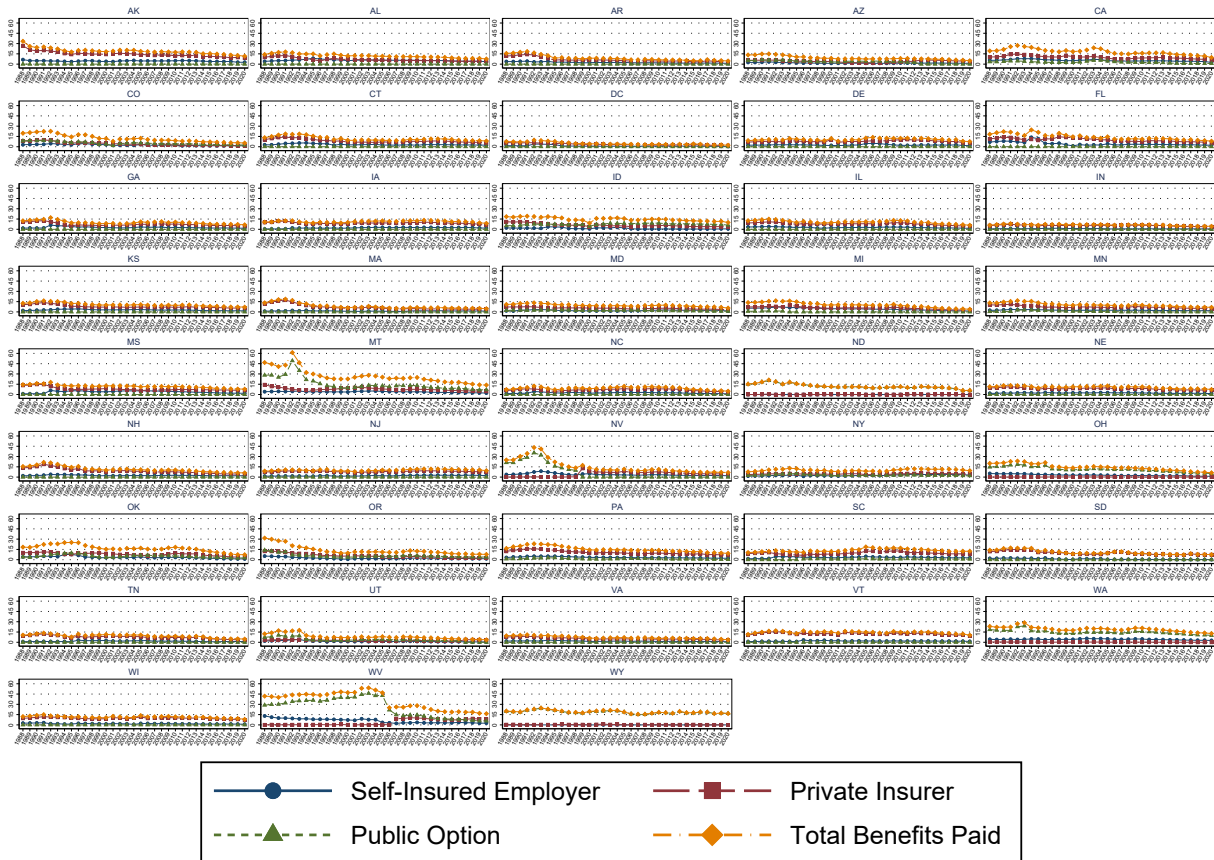
Notes: This figure displays average losses incurred and premiums earned in private market states - in panels (a) and (c) - and new public option states - in panels (b) and (d) - across calendar time (year) and event time, respectively. A linear trend line is drawn through all points in panels (a) and (c) and drawn separately in panels (b) and (d) for public option states before and after formation (at time = 0). Vertical lines in the left-side graphs depict public option creation years. Sources: NAIC 1988-2020.

Figure A.4: Trends in Benefits Paid, New Public Option States



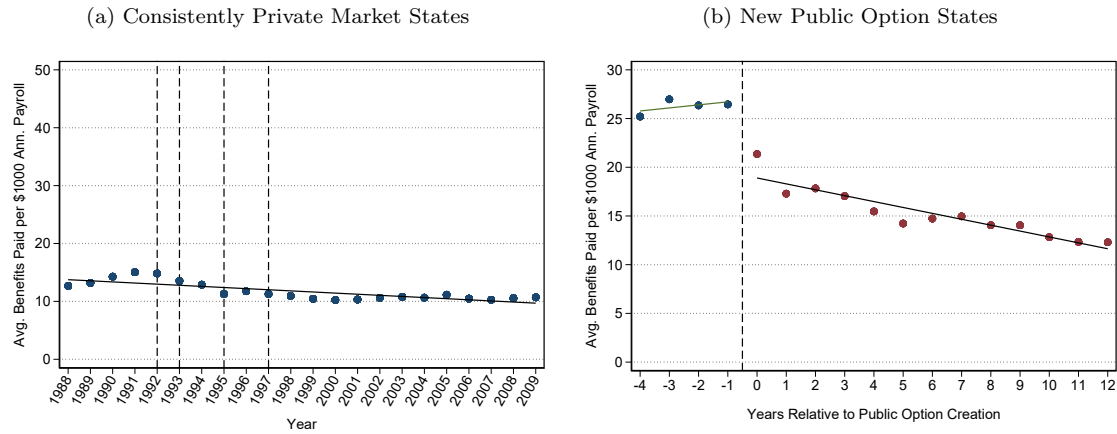
Notes: This figure displays workers' compensation benefits paid per \$1,000 of annual payroll within each state over time. Benefits paid by self-insured employers are given by the blue (circle) lines; private insurer benefits paid are given by the red (square) lines. Public option benefits paid are given by the green (triangle) lines, and total benefits paid across all sources are given by the orange (diamond) circles. Sources: CBP, SSB, and the National Academy of Social Insurance 1988-2020.

Figure A.5: Trends in Benefits Paid, All Other States



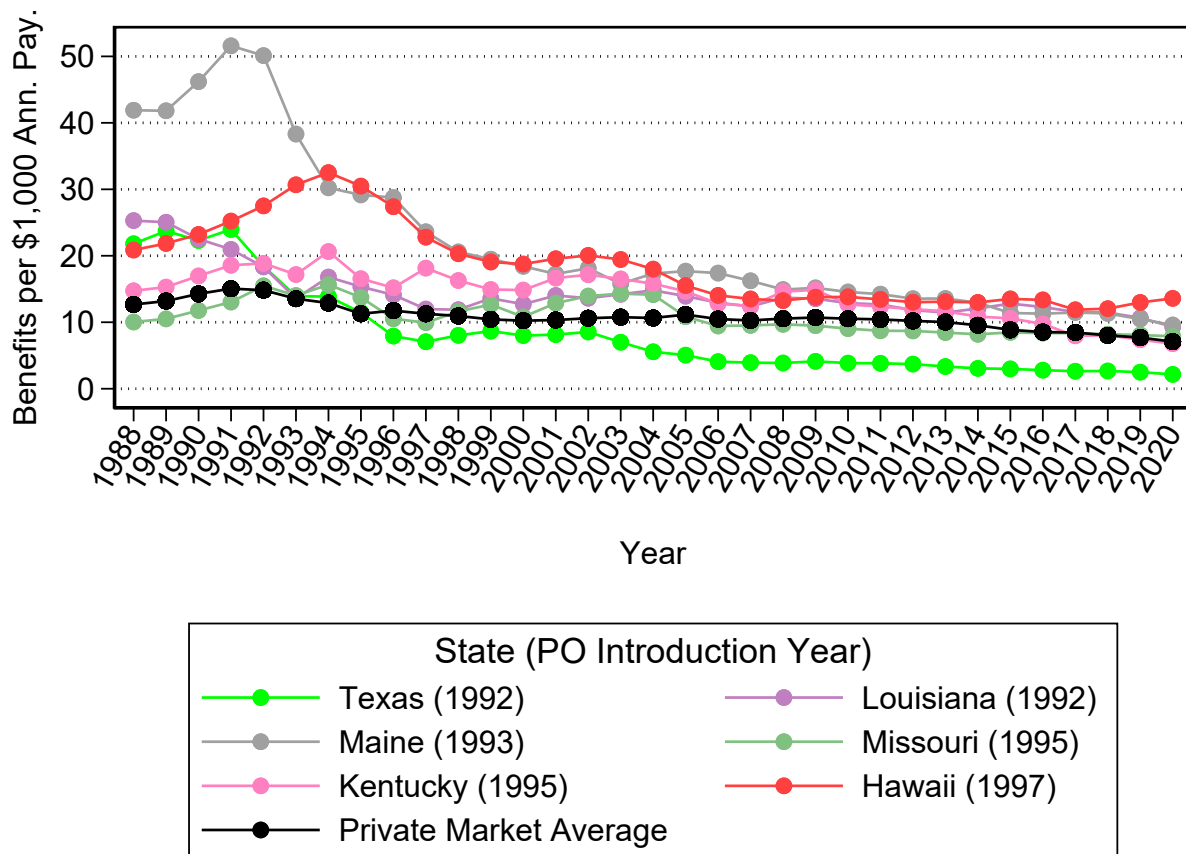
Notes: This figure displays workers' compensation benefits paid per \$1,000 of annual payroll within each state over time. Benefits paid by self-insured employers are given by the blue (circle) lines; private insurer benefits paid are given by the red (square) lines. Public option benefits paid are given by the green (triangle) lines, and total benefits paid across all sources are given by the orange (diamond) circles. Sources: CBP, SSB, and the National Academy of Social Insurance 1988-2020.

Figure A.6: Trends in Benefits Paid - Means



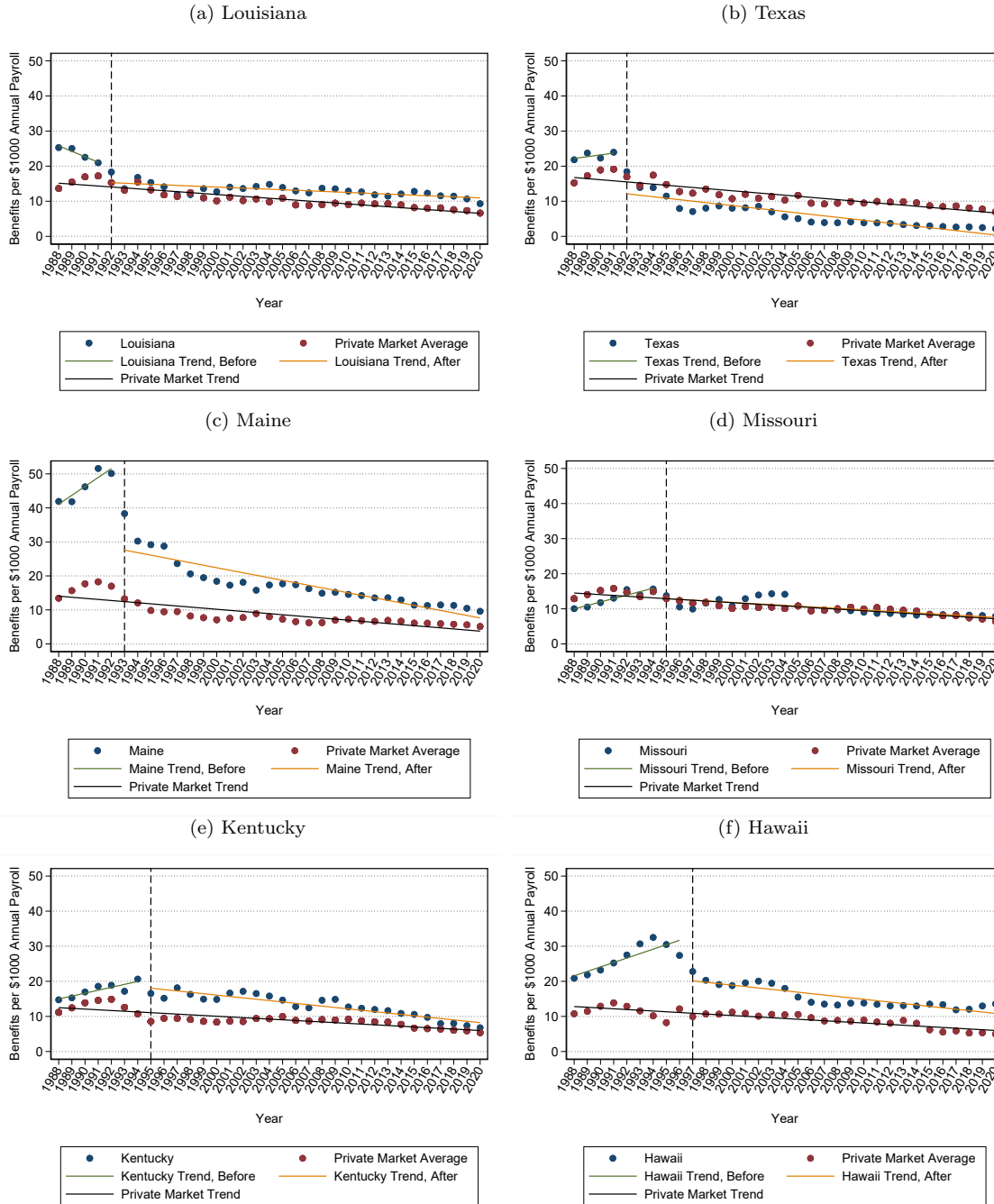
Notes: This figure displays mean workers' compensation benefits paid per \$1,000 of annual payroll within consistent private market states in panel (a) and new public option states in panel (b), across calendar time (year) and event time, respectively. A linear trend line is drawn through all points in panel (a) and drawn separately in panel (b) before and after formation (at time = 0). Vertical lines in panel (a) indicate public option creation years. Sources: CBP, SSB, and the National Academy of Social Insurance 1988-2020.

Figure A.7: Trends in Benefits Paid, Public Option States vs. Private Market Average



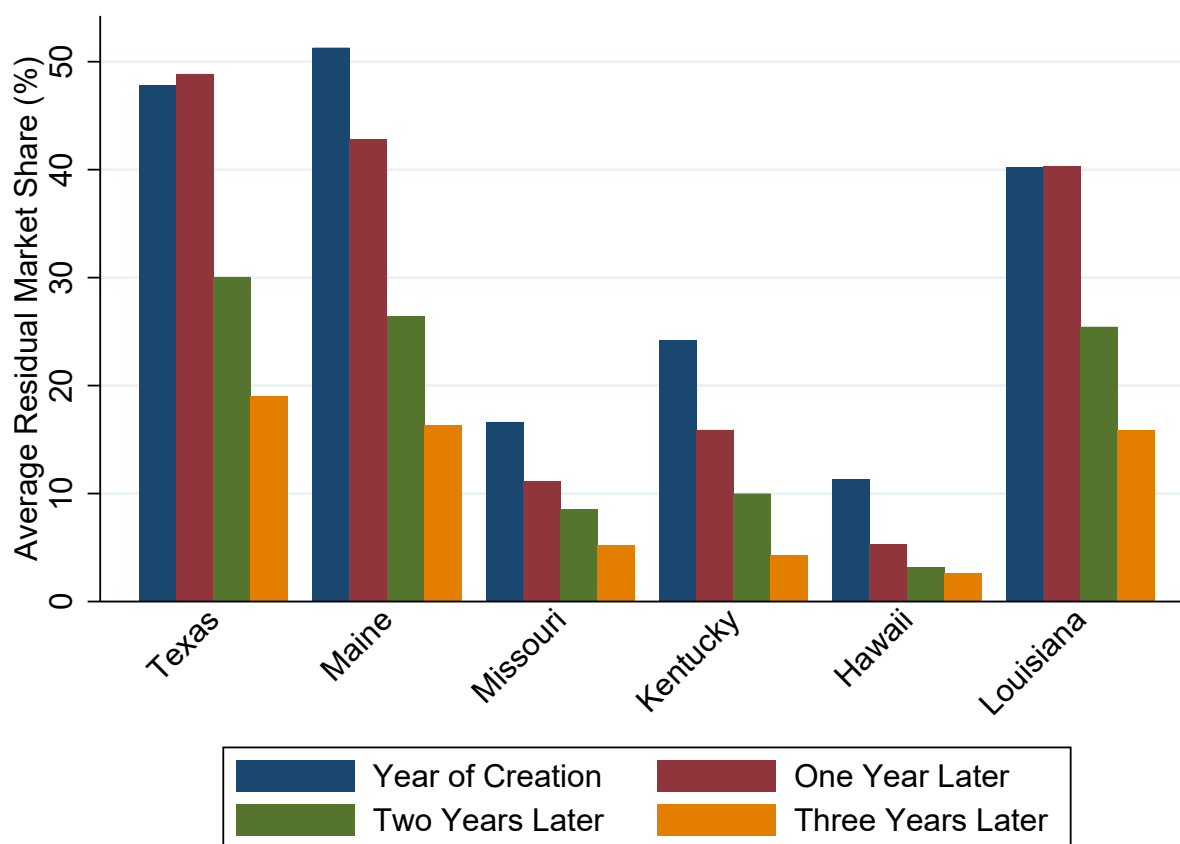
Notes: This figure displays workers' compensation benefits paid per \$1,000 of annual payroll within each new public option state over time, compared to the average among the consistently private market states. Sources: CBP, SSB, and the National Academy of Social Insurance 1988-2020.

Figure A.8: Trends in Benefits Paid, Post-Match



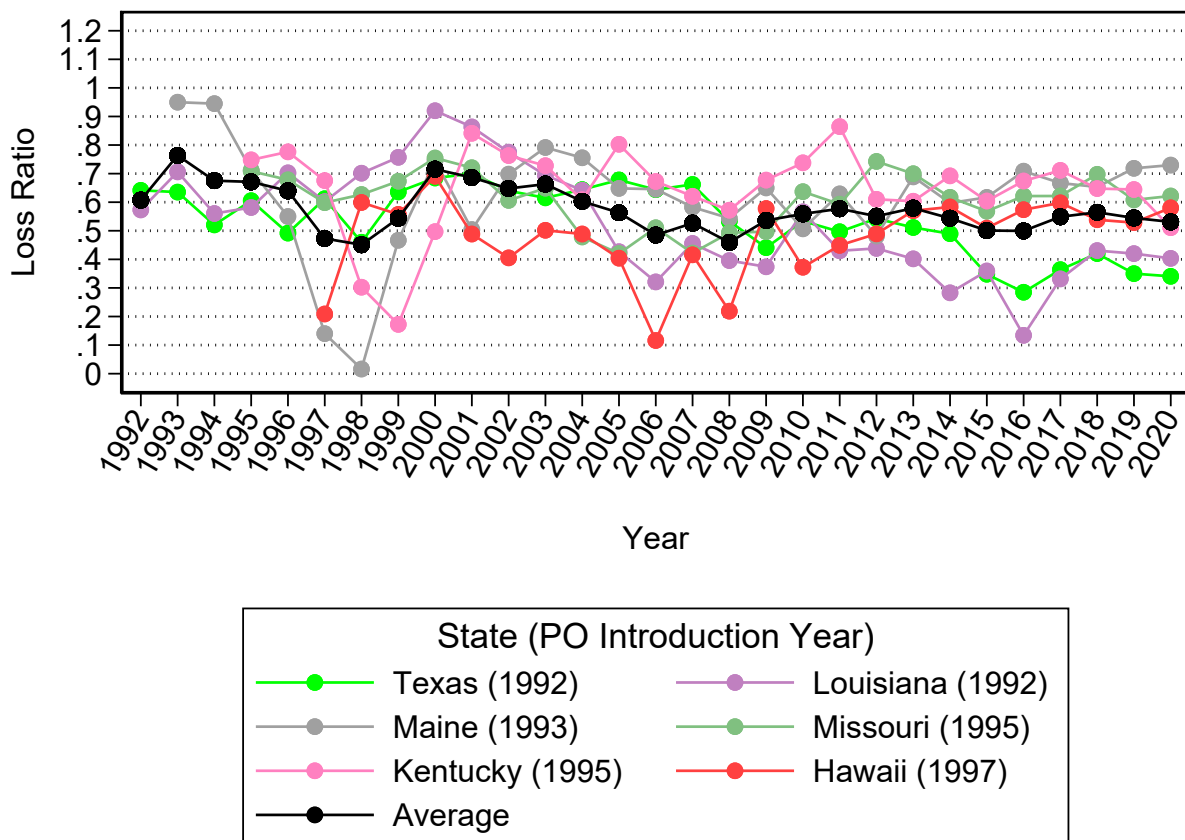
Notes: This figure displays benefits paid levels per \$1,000 of annual payroll in each public option state (given by panel title) compared to its matched comparison states from the private-only subset. Blue dots represent benefits paid in the public option state, with a green trend line before formation and an orange trend line after formation. Red dots represent the matched comparator average benefits paid levels, connected with a black trend line. Sources: CBP, SSB, and the National Academy of Social Insurance 1988-2020.

Figure A.9: Private-Only Residual Markets Before and After Public Option Formation



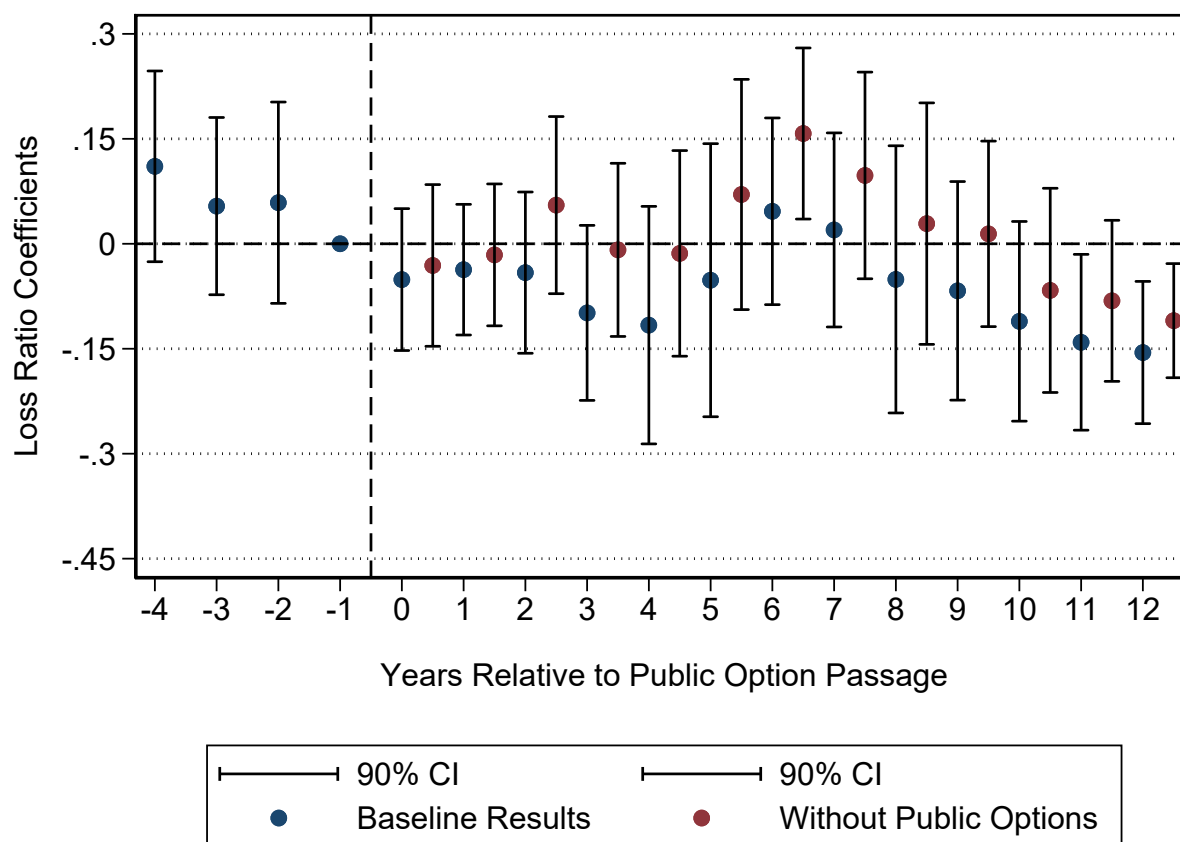
Notes: This figure displays residual market shares in consistently in private market states matched to each new public option state (given by x-axis labels) the year the public option launched, and then for each of the three years after that. Only two matched private market states aren't included in their complete form: Wisconsin and Florida (after 1995). Sources: [Thomason et al. \(2001\)](#), [Buhro \(2011\)](#), and [NCCI \(2000\)](#).

Figure A.10: Public Option Loss Ratios Over Time



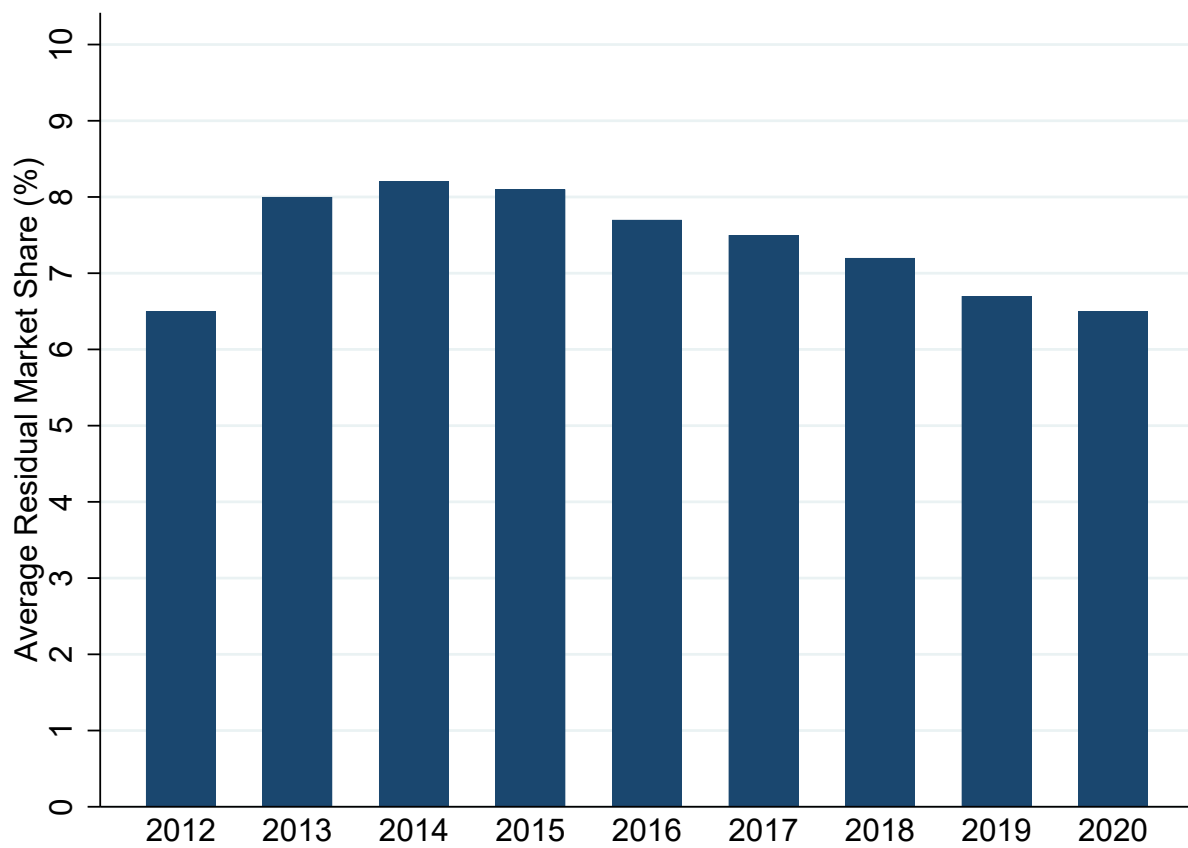
Notes: This figure displays the loss ratios of each public option over time, along with the average among the public options. Sources: NAIC 1992-2020.

Figure A.11: Loss Ratio Results With(out) Public Options



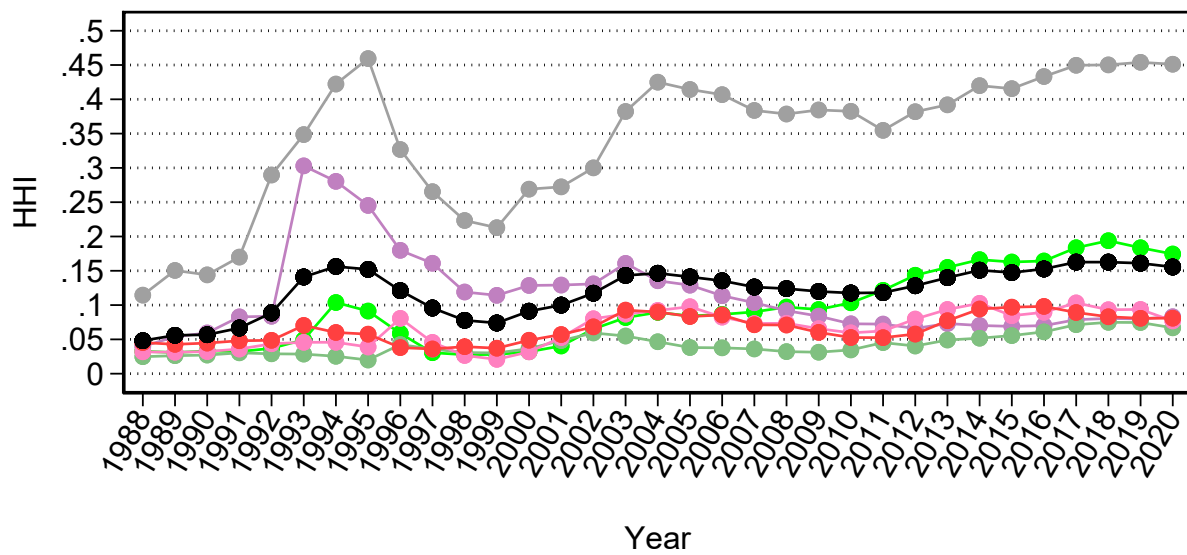
Notes: This figure displays point estimates from estimating Equation 1 with respect to loss ratios. 90% confidence intervals are used. Blue dots duplicate panel (b) of Figure 10, while red dots repeat that analysis after dropping the public options, which only affects the period after formation (time ≥ 0). Sources: NAIC 1988-2020.

Figure A.12: Recent Residual Market Shares



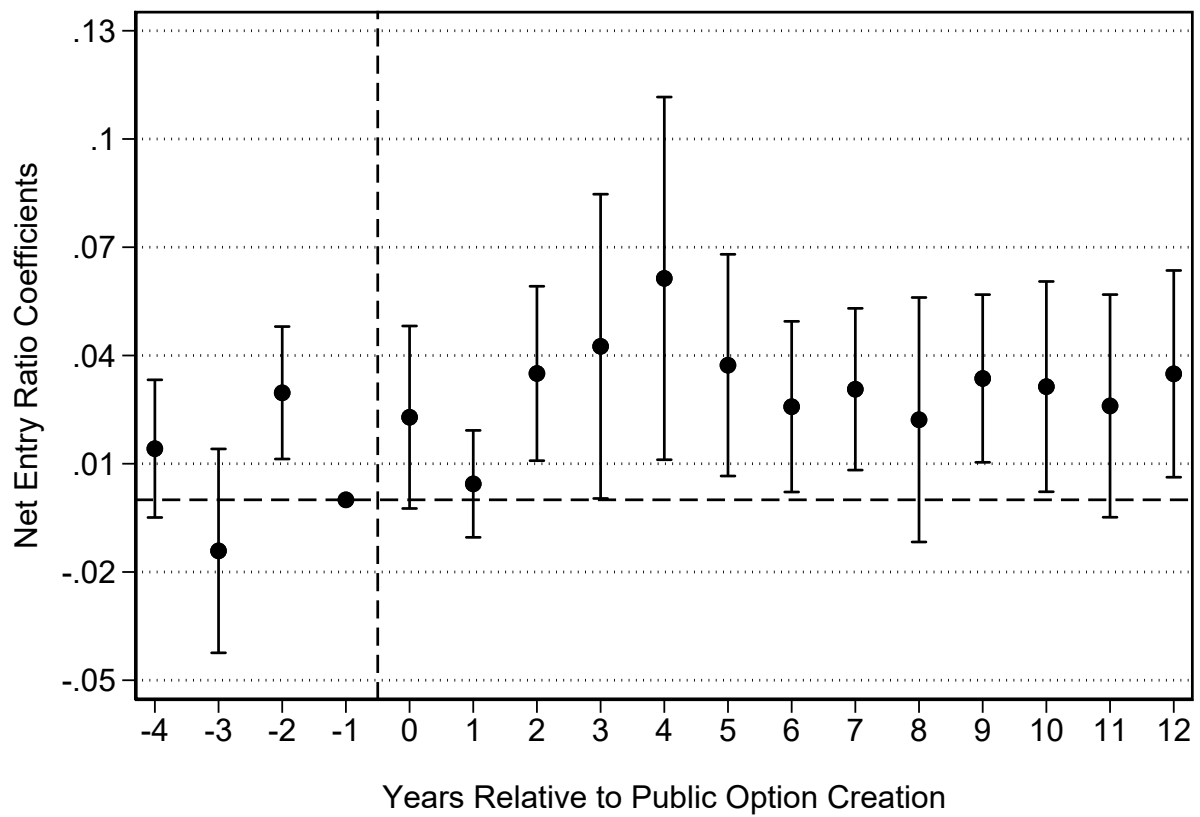
Notes: This figure displays the percentage of direct premiums written by the residual market in NCCI states over time. Sources: [NCCI \(2016\)](#) and [NCCI \(2020\)](#).

Figure A.13: Public Option States - HHIs Over Time



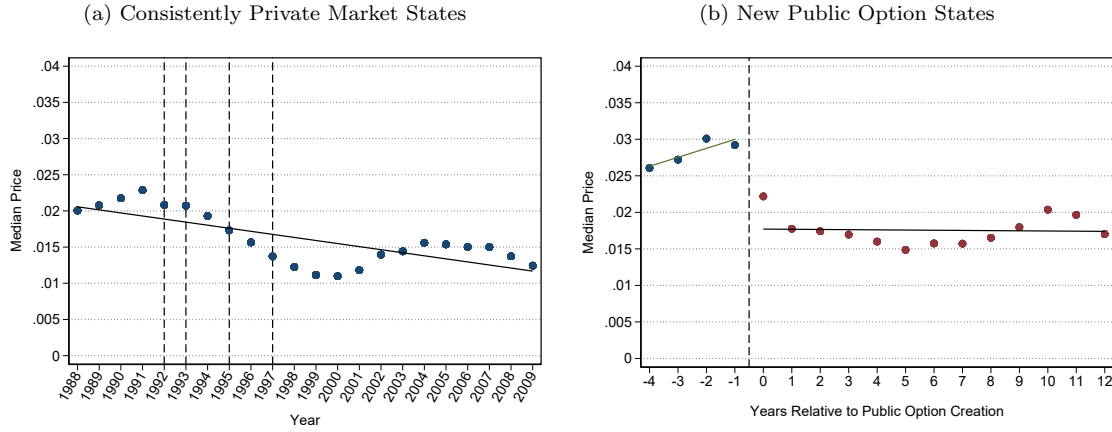
Notes: This figure displays HHI measures in each public option state, and the running average, across time.
Sources: NAIC 1988-2020.

Figure A.14: Net Entry Ratio



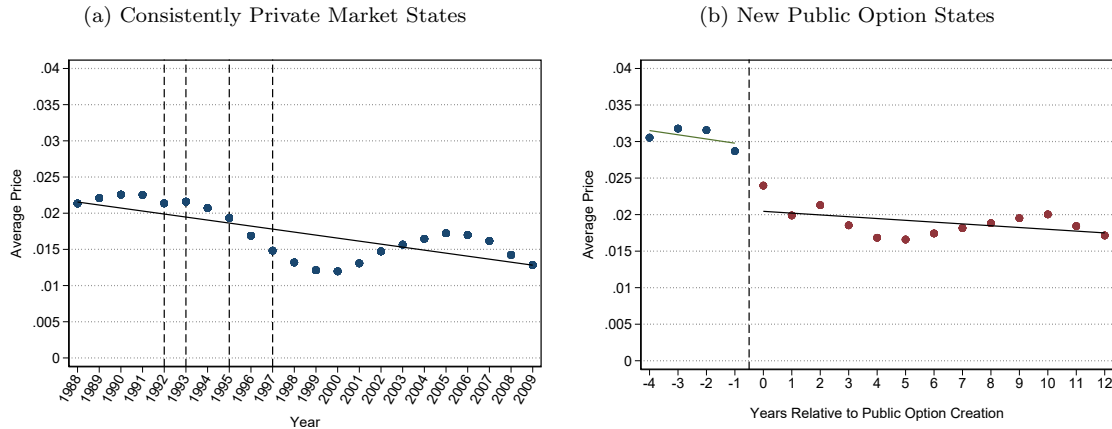
Notes: This figure displays point estimates from estimating Equation 1 with respect to the net entry ratio. 90% confidence intervals are used. Sources: NAIC 1988-2020.

Figure A.15: Trends in Price - Medians



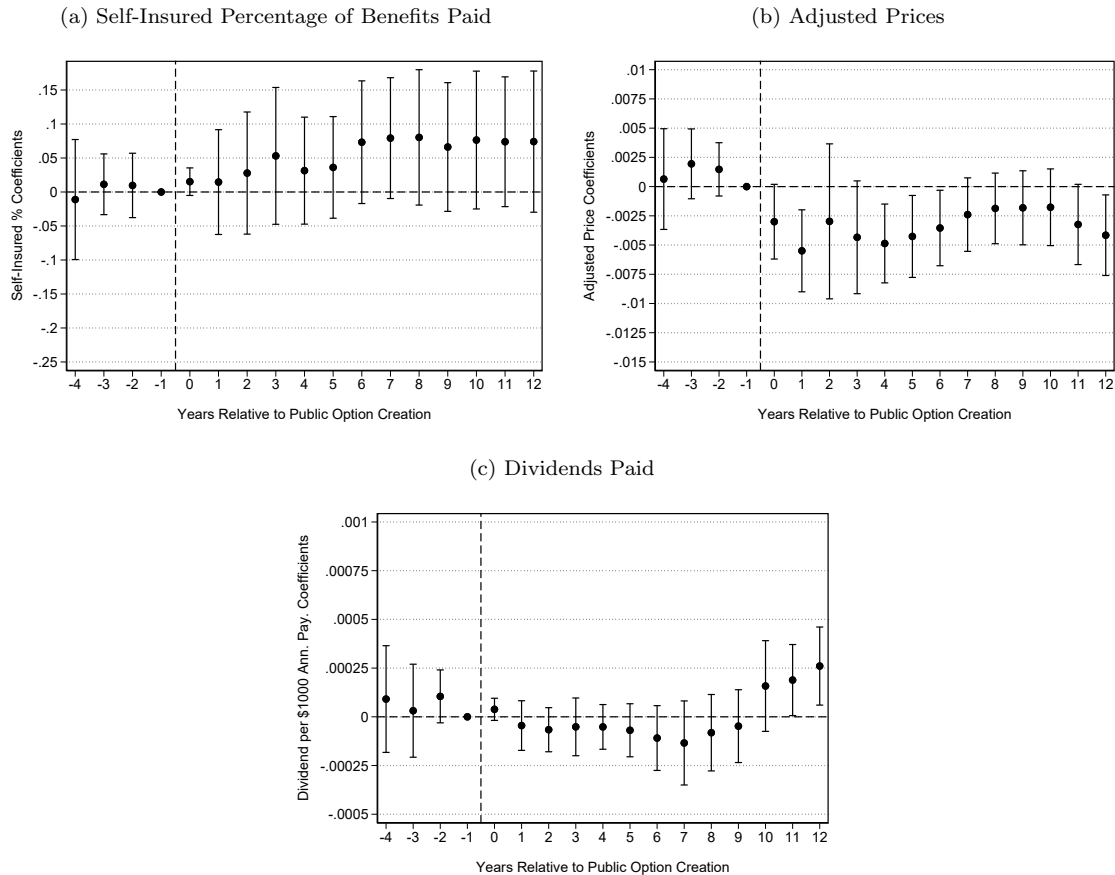
Notes: This figure displays median workers' compensation insurance price per \$1 of annual payroll within consistent private market states in panel (a) and new public option states in panel (b), across calendar time (year) and event time, respectively. A linear trend line is drawn through all points in panel (a) and drawn separately in panel (b) before and after formation (at time = 0). Vertical lines in panel (a) indicate public option creation years. Sources: NAIC, CBP, SSB, and the National Academy of Social Insurance 1988-2020.

Figure A.16: Trends in Price - Means



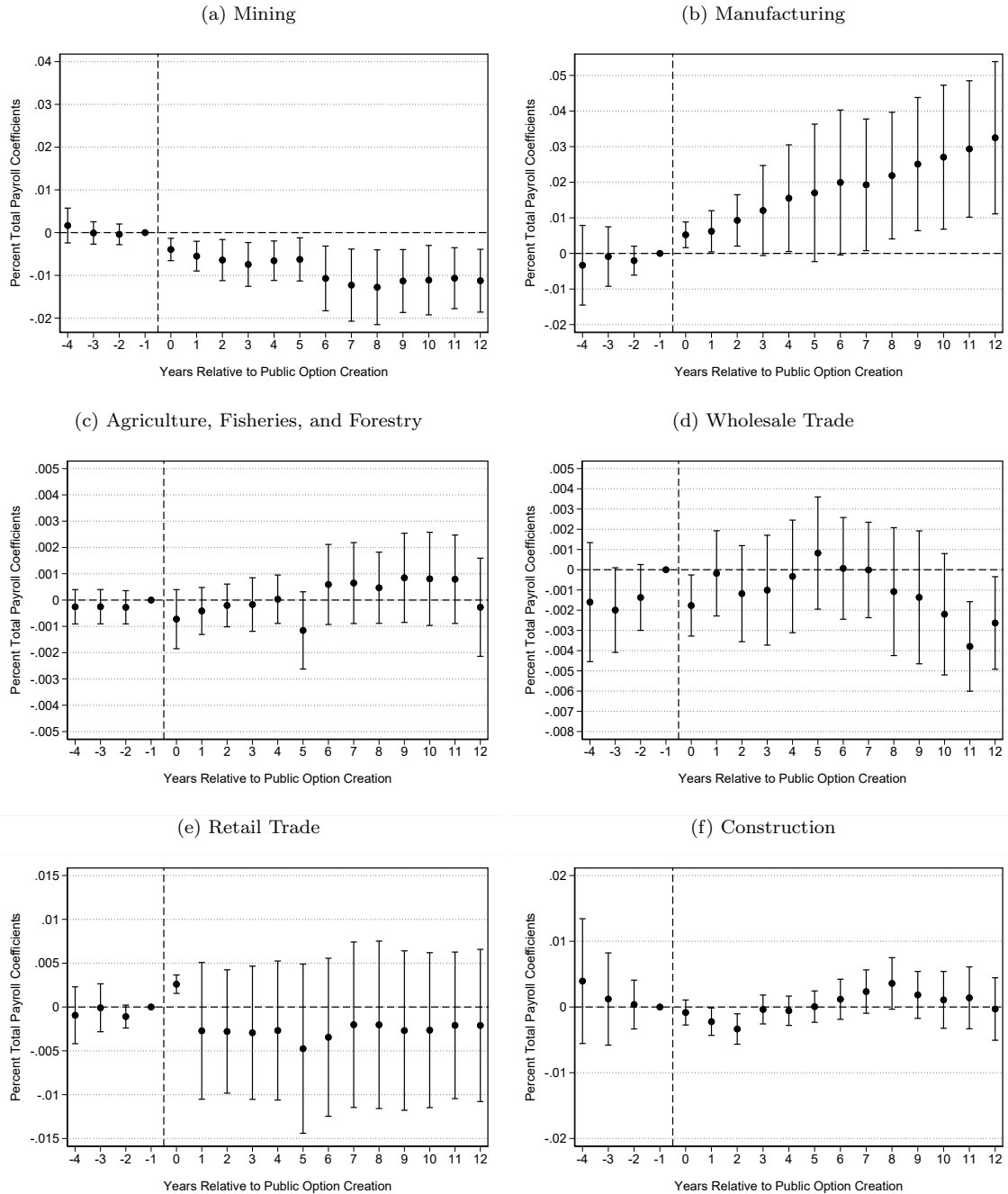
Notes: This figure displays mean workers' compensation insurance price per \$1 of annual payroll within consistent private market states in panel (a) and new public option states in panel (b), across calendar time (year) and event time, respectively. A linear trend line is drawn through all points in panel (a) and drawn separately in panel (b) before and after formation (at time = 0). Vertical lines in panel (a) indicate public option creation years. Sources: NAIC, CBP, SSB, and the National Academy of Social Insurance 1988-2020.

Figure A.17: Stacked Event Study Results - Self-Insured Benefits, Adjusted Price, Dividends



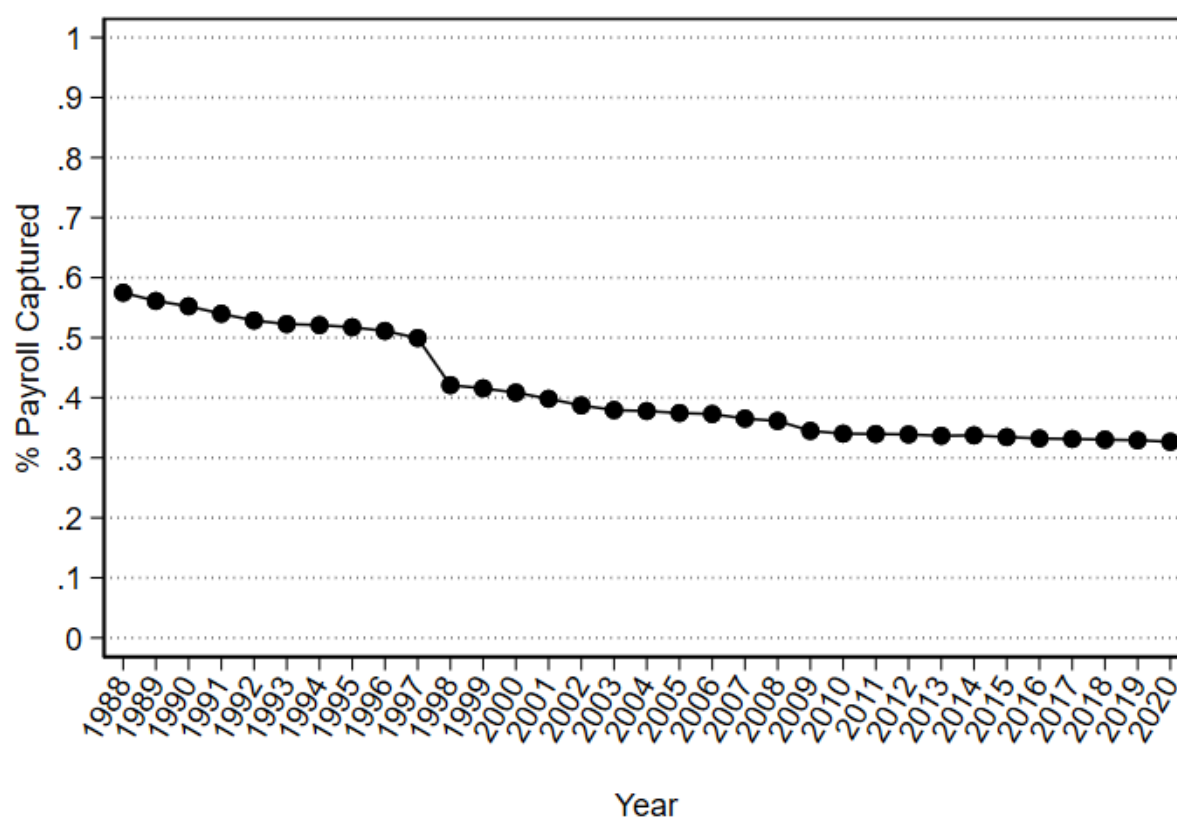
Notes: This figure displays point estimates from estimating Equation 1 with three outcomes: self-insured percentage of total benefits paid in panel (a), prices adjusted for dividends paid in panel (b), and then dividends paid per \$1,000 of annual payroll in panel (c). 90% confidence intervals are used. Note the different axis values, given the vastly different outcomes being graphed in each. Sources: NAIC, CBP, SSB, and the National Academy of Social Insurance 1988-2020.

Figure A.18: Stacked Event Study Results - Industry Mix



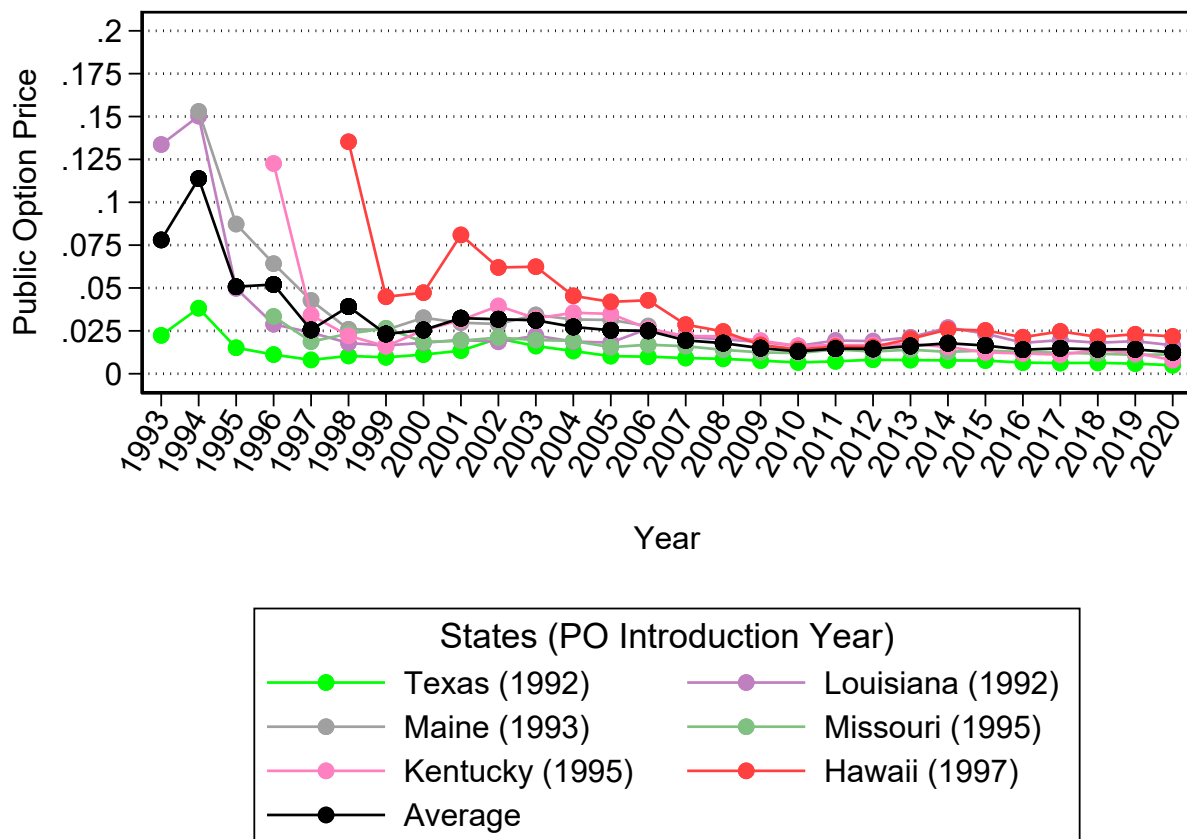
Notes: This figure displays point estimates from estimating Equation 1 with the outcome as the percentage of a state's total payroll coming from the industry named in each sub-figure title. 90% confidence intervals are used. Note the different axis values, given the vastly different baselines in each industry. Sources: CBP 1988-2020.

Figure A.19: Payroll Captured



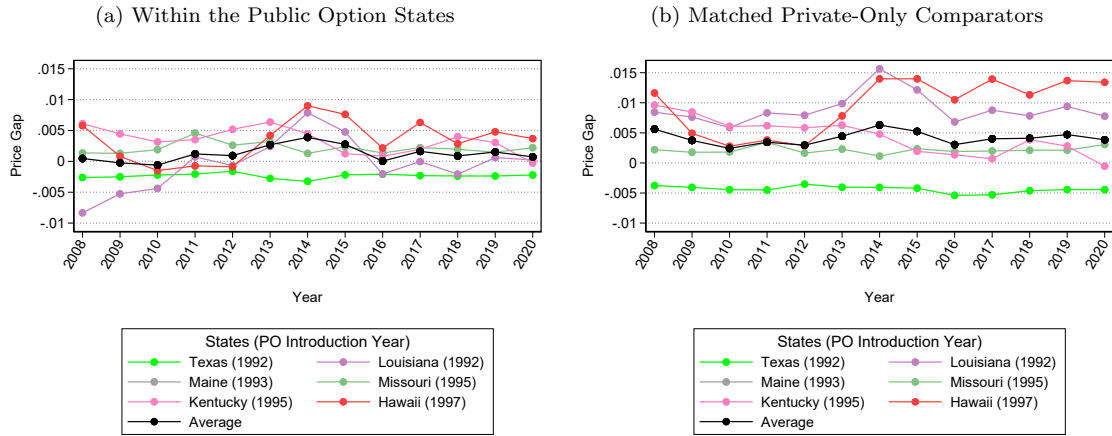
Notes: This figure shows how much of total payroll, on average, is captured by the six broad categories from Appendix Figure A.18, over time. Sources: CBP 1988-2020.

Figure A.20: Public Option Prices



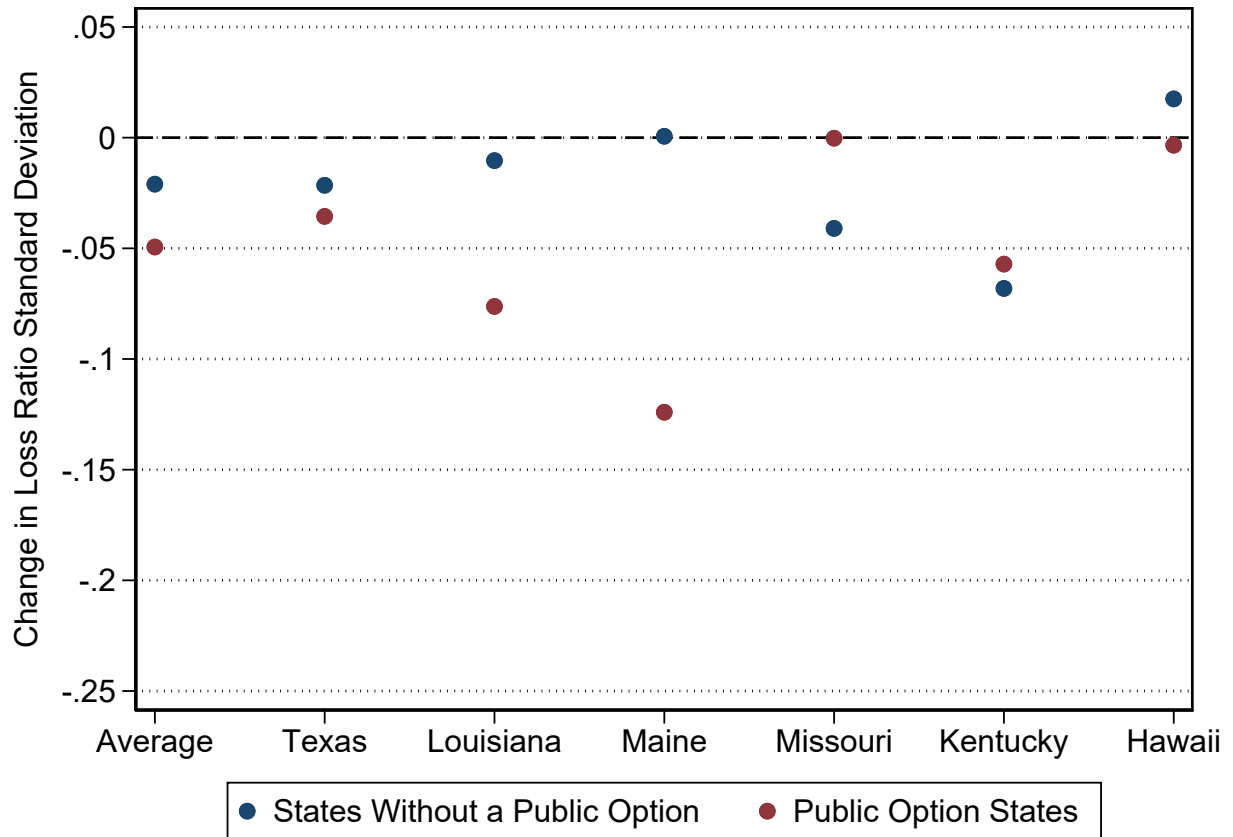
Notes: This figure displays workers' compensation insurance price per \$1 of annual payroll for public options over time, along with the average public option price among the six new public option states. Sources: NAIC, CBP, SSB, and the National Academy of Social Insurance 1993-2020.

Figure A.21: Price Gaps, 2008-2020 - Public Options vs. the Private Market



Notes: This figure displays the gaps between the public option price and the private market price over time, specific to the 2008-2020 period. The private market is defined as all private insurers in the public option's state in panel (a) and the matched group of private-only comparators in panel (b). Sources: NAIC, CBP, SSB, and the National Academy of Social Insurance 2008-2020.

Figure A.22: Firm-Level Stability in Loss Ratios



Notes: This figure displays the change in the standard deviation of firm-level loss ratios between the three years before public option formation and then the three years after (including the year of formation) among private insurers in matched private-only states (blue dots) and states with a new public option (red dots). States are grouped by their reference public option (x-axis), along with the average of the gaps across all six categories on the far left. This graph drops firm-level loss ratios outside of the 10/90 percentiles. Sources: NAIC 1989-1999.